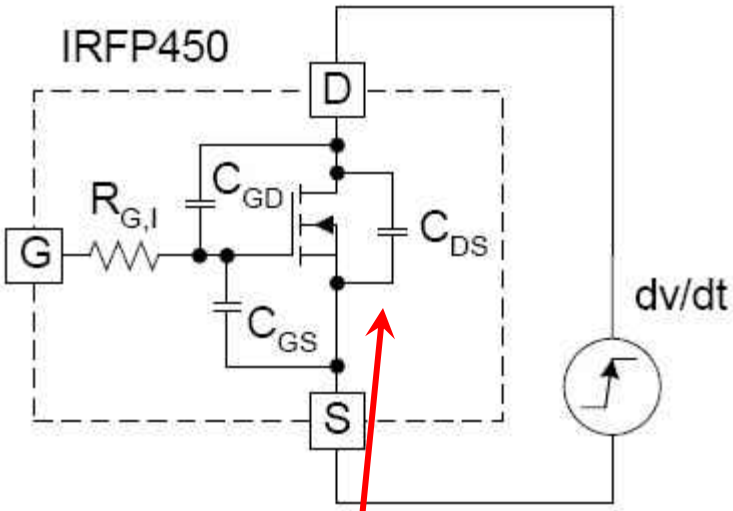
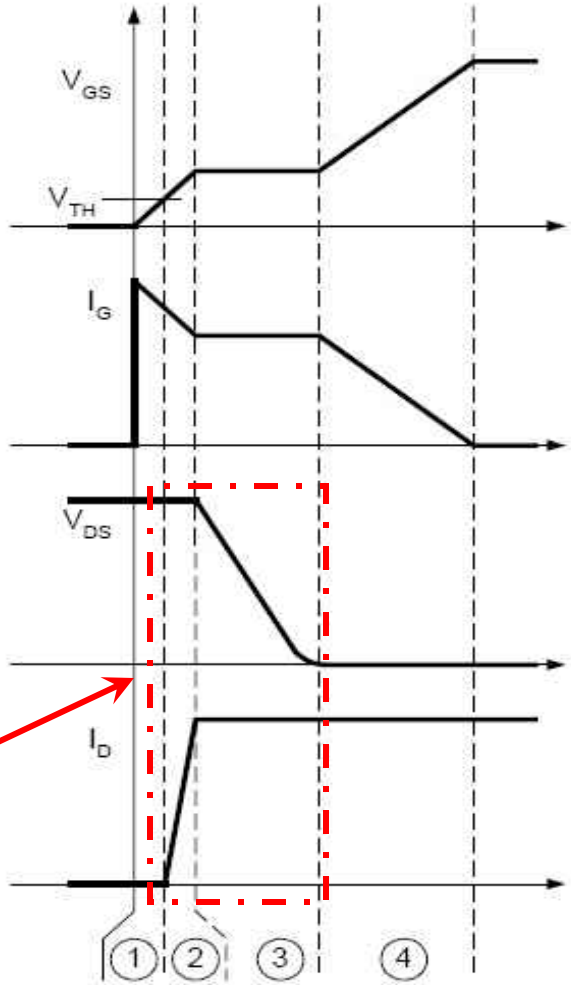
LLC设计步骤

- LLC的基本原理
- LLC的设计方法
- LLC的几个问题

LLC的基本原理

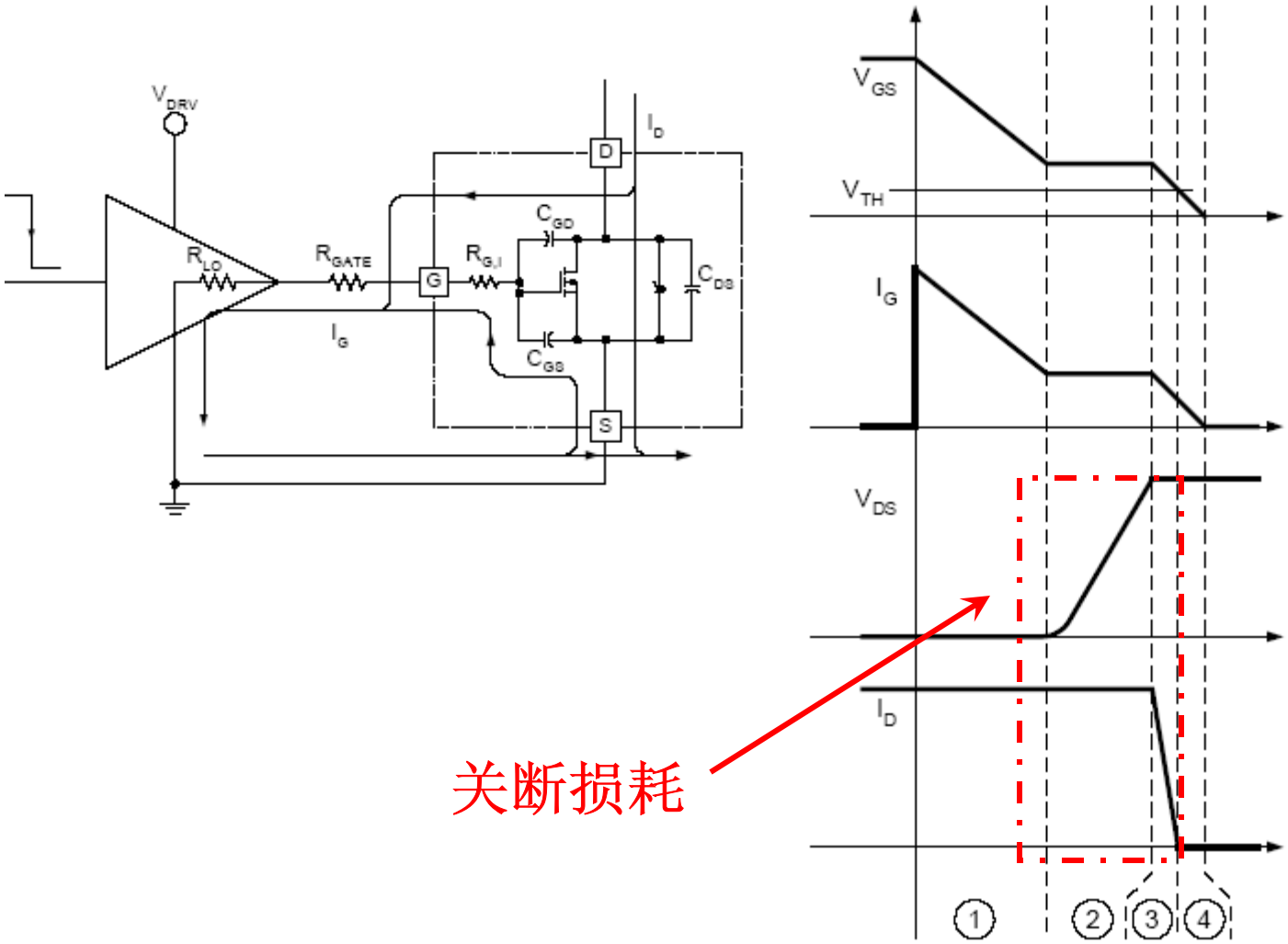
MOSFET适合零电压开关



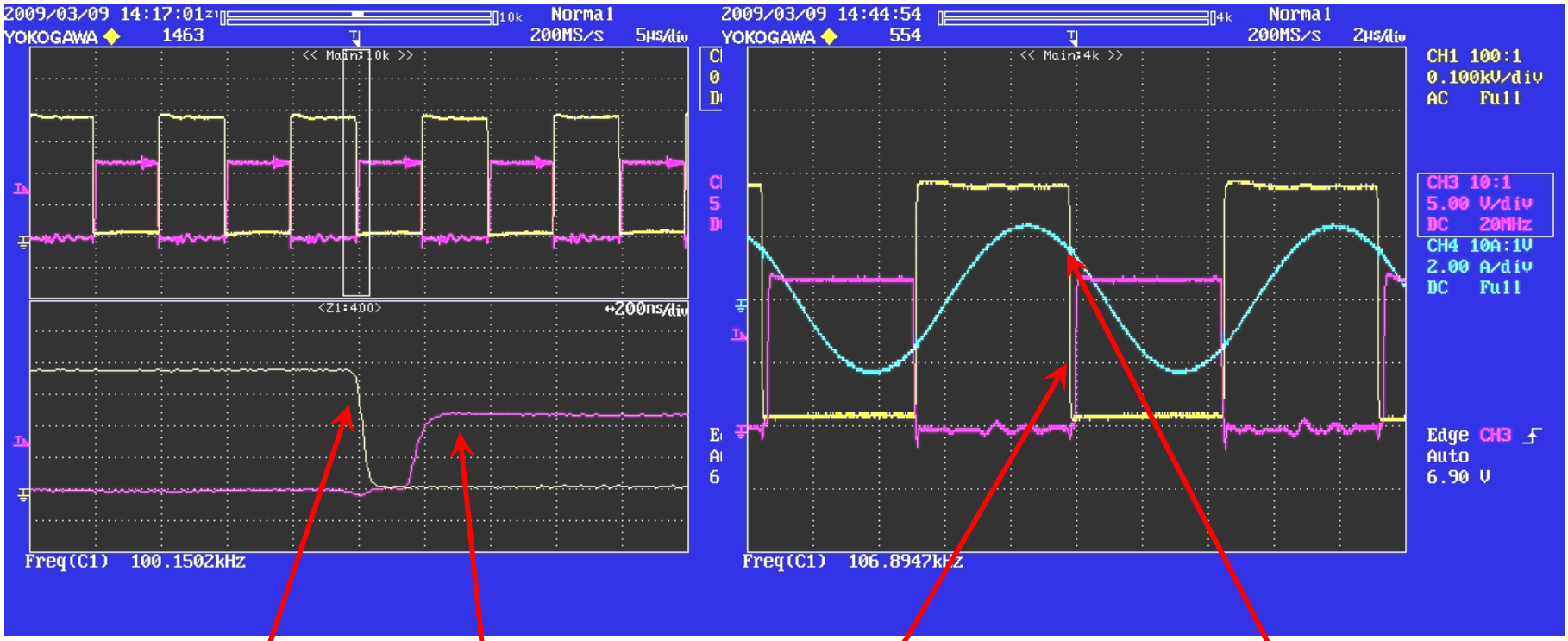


$$\frac{1}{2}CV_{DS}^2 \quad + \quad \text{开通损耗}$$

MOSFET适合零电压开关



MOSFET的零电压开关



V_{DS}

V_{GS}

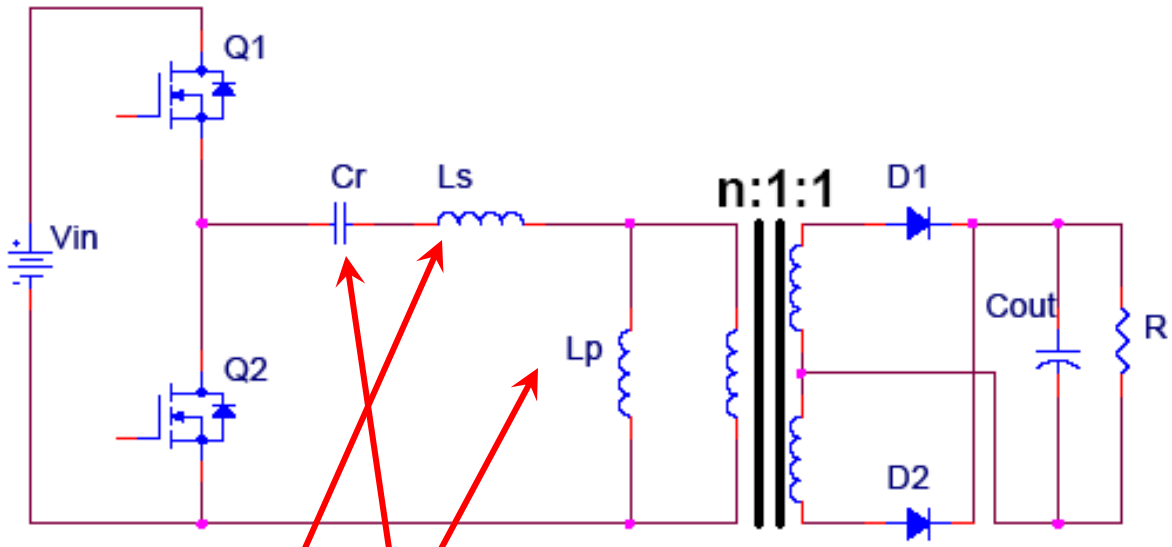
i_r

关断损耗避免不了

MOSFET开通前,其Vds电压已经为零,则为零电压开通(ZVS)

ZVS的实现需要驱动信号来时有电流从S到D流通,LLC可以实现

LLC的架构



$$k = \frac{L_p}{L_s}$$

$$f_r = \frac{1}{2\pi\sqrt{L_s C_r}}$$

$$f_{r2} = \frac{1}{2\pi\sqrt{(L_s + L_p)C_r}}$$

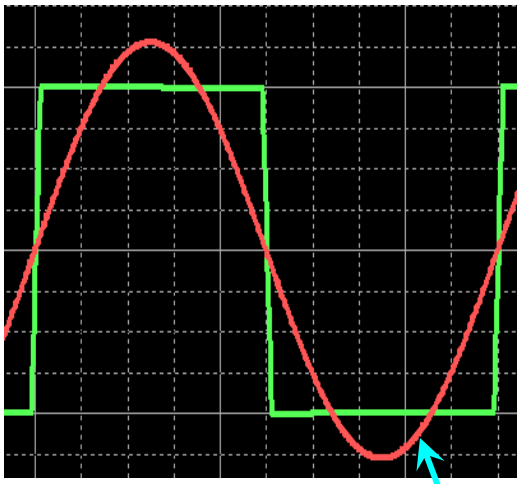
L L C

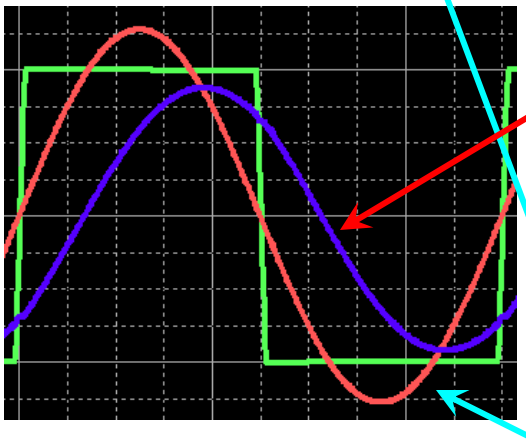
两个谐振频率

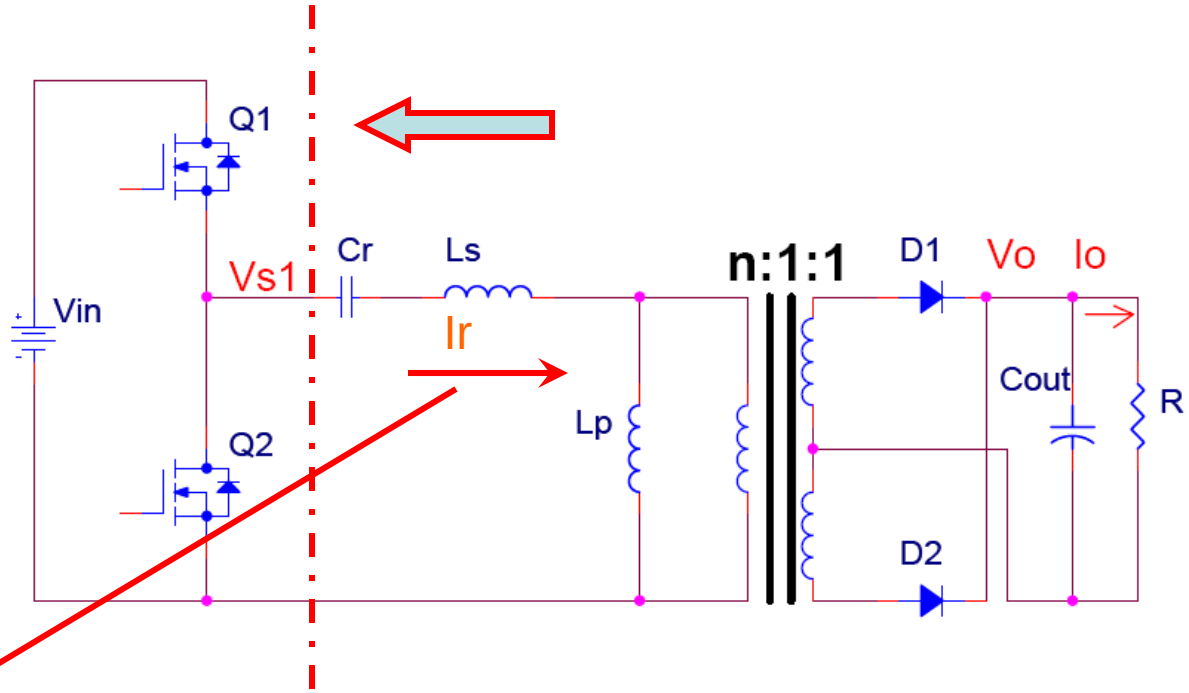
LLC的详细工作过程

Acrobat Document

输入FHA等效电路



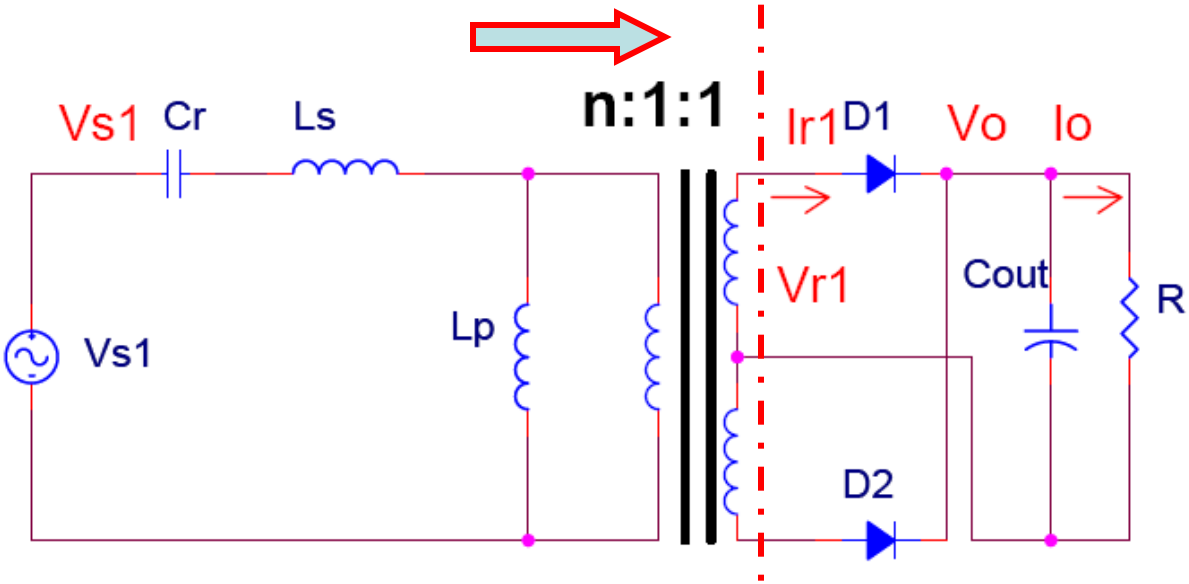


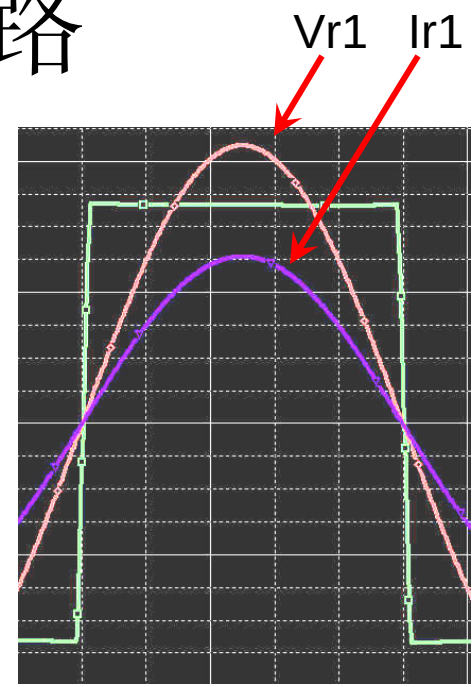


$$Vs = \frac{Vin}{2} + \frac{2}{\pi} Vin \sum_{n=1,3,5\ldots} \frac{1}{n} \sin(2n\pi f_{sw} t)$$

$$Vs1 = \frac{2}{\pi} Vin \sin(2\pi f_s t)$$

输出FHA等效电路





$$Vr1 = \frac{4}{\pi} V_o \sin(2\pi f_s t - \varphi_R)$$

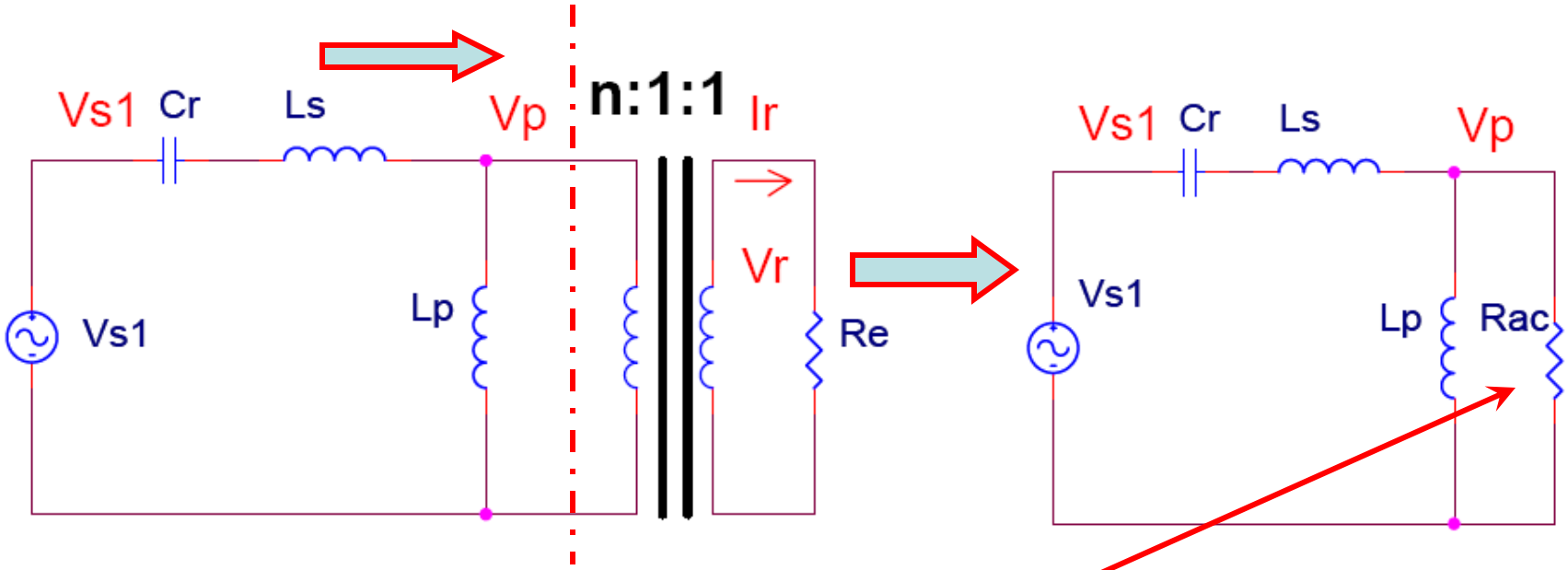
$$Ir1 = I_{R1} \sin(2\pi f_s t - \varphi_R)$$

$$I_o = \frac{2}{Ts} \int_0^{Ts/2} I_{R1} \left| \sin(2\pi f_s t - \varphi_R) \right| dt = \frac{2}{\pi} I_{R1}$$

Vr1, Ir1同相,所以阻抗为电阻 →

$$\mathrm{Re} = \frac{V_{r1}(t)}{i_{r1}(t)} = \frac{8}{\pi^2} * \frac{V_o}{I_o} = \frac{8R}{\pi^2}$$

输出FHA等效电路

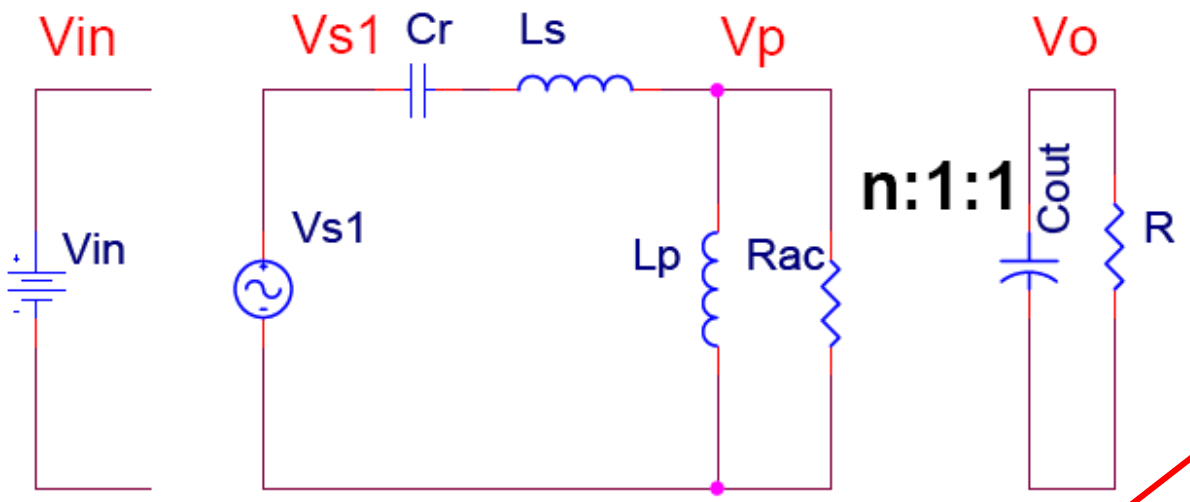


等效到初级侧

$$R_{AC} = n^2 \mathrm{Re} = n^2 \frac{8}{\pi^2} R$$

$$V_p = nV_{r1} = n\frac{4}{\pi}V_o$$

LLC的简化FHA等效电路



归一化增益

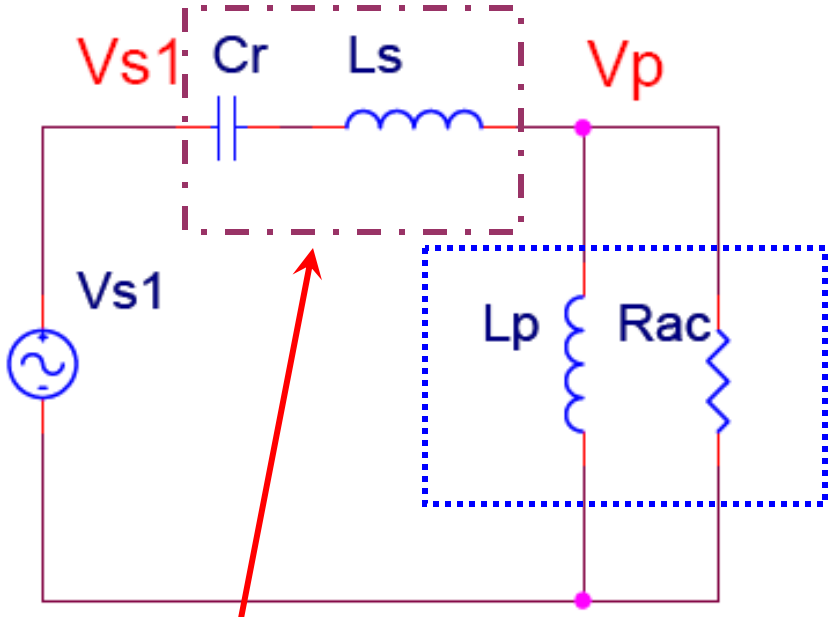
$$Vs1 = \frac{2}{\pi} Vin \Rightarrow Vin = \frac{\pi}{2} Vs1$$

$$V_p = nV_{r1} = n\frac{4}{\pi} V_o \Rightarrow V_o = \frac{\pi}{4n} V_p$$

$$|G| = \frac{nVo}{\frac{Vin}{2}} = \frac{n\frac{\pi}{4n}V_P}{\frac{\pi}{4}V_{S1}} = \frac{V_P}{V_{S1}}$$

$$= \frac{sLp // Rac}{1/sCr + sLr + (sLp // Rac)}$$

LLC的稳压原理



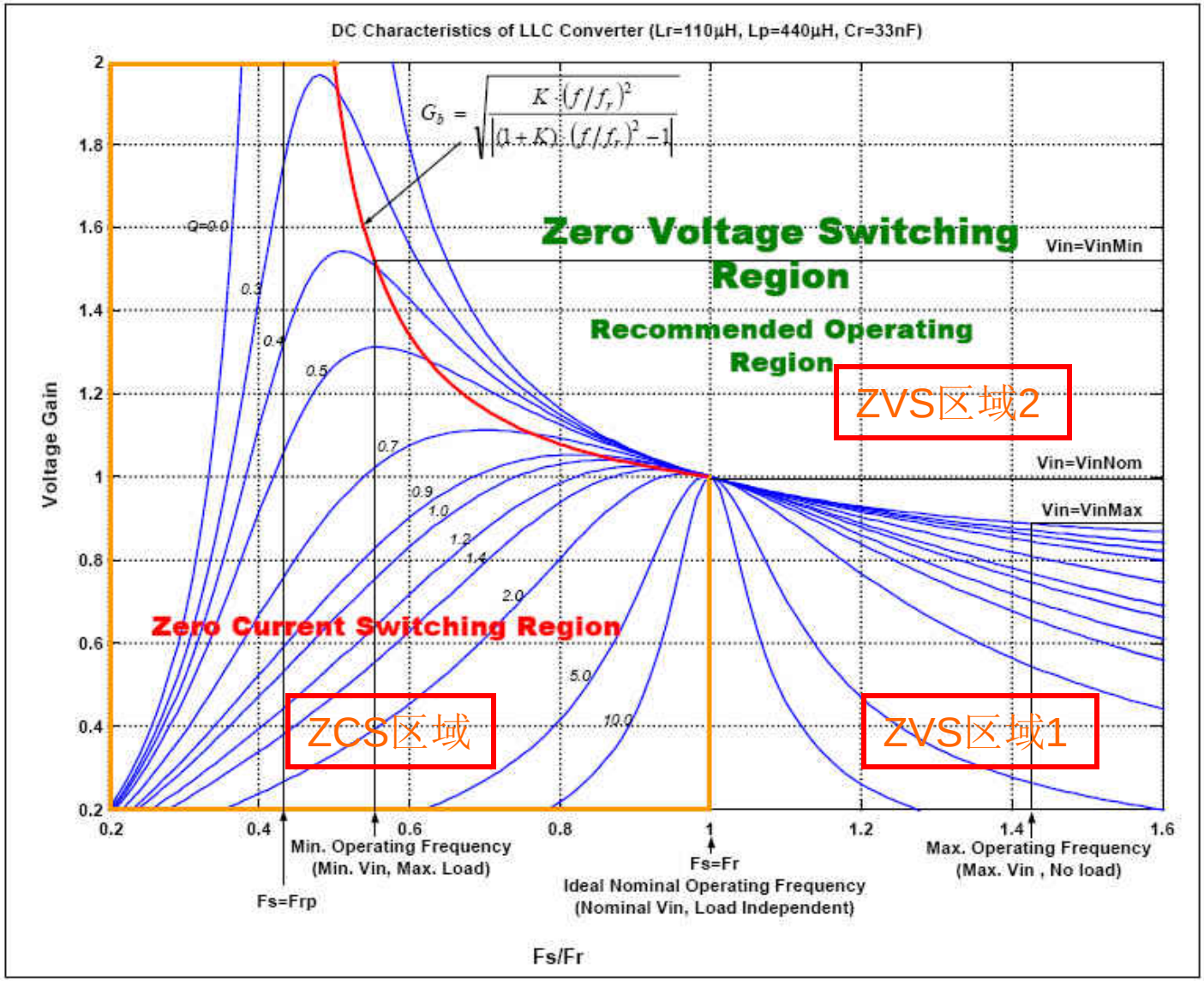
输入或负载变化时引起Vp变化

$$k = \frac{L_p}{L_s} \qquad x = \frac{f}{fr} \qquad Q = \frac{2\pi f_r L_s}{R_{AC}}$$

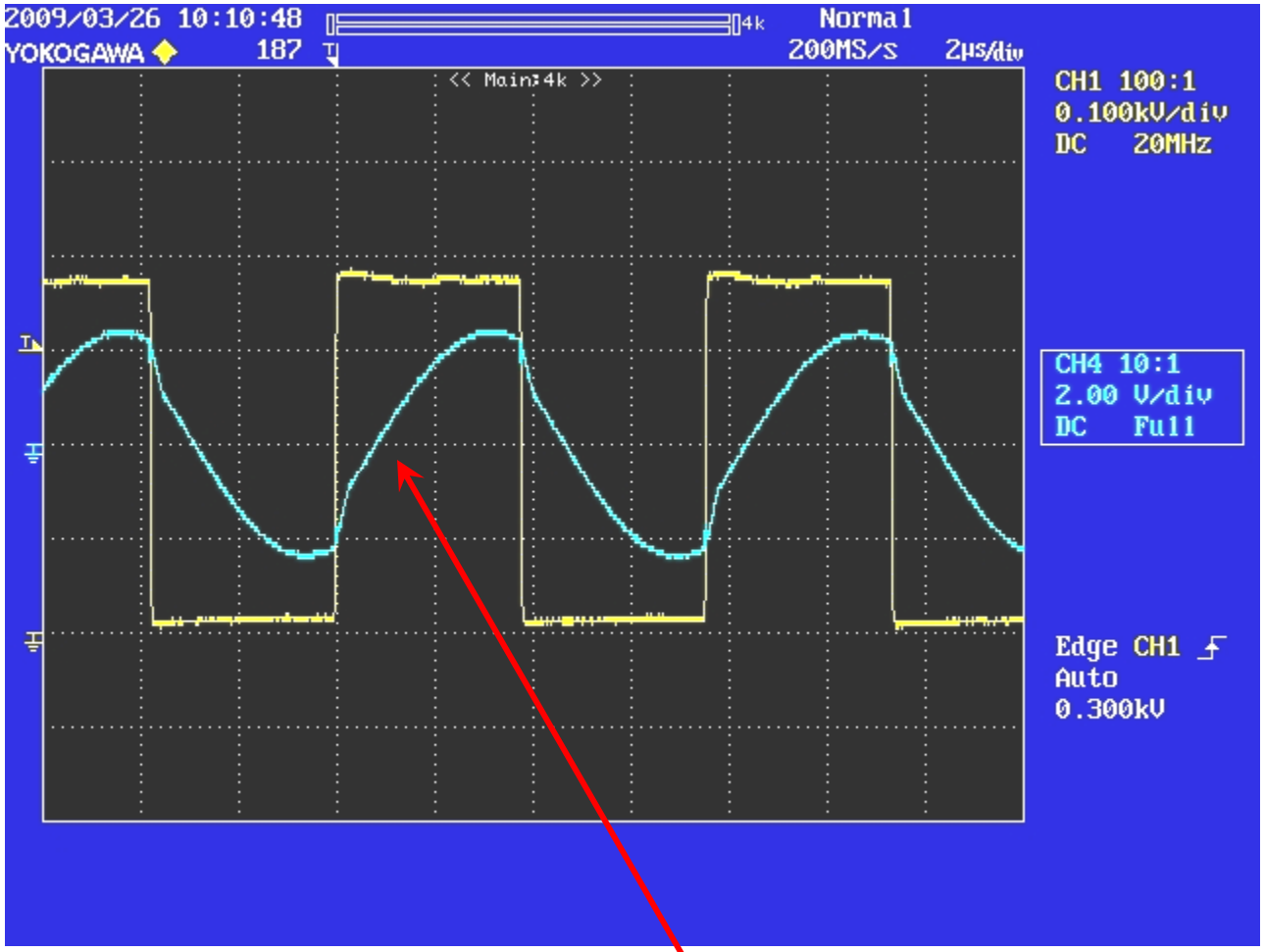
通过改变频率使1/sCr+sLr的分压相应改变，最终维持负载电压不变,即Vp不变

$$|G(f)| = \frac{k*(x)^2}{\sqrt{\{(1+k)*(x)^2-1\}^2+\{Qk(x)*[(x)^2-1]\}^2}}$$

LLC的DC特性


DC Characteristics of LLC Converter (Lr=110μH, Lp=440μH, Cr=33nF)
$G_b = \sqrt{\frac{K \cdot (f/f_r)^2}{\sqrt{|(1+K)\cdot(f/f_r)^2 - 1|}}}$
Zero Voltage Switching Region
Recommended Operating Region
ZVS区域2
Vin=VinMin
Vin=VinNom
Vin=VinMax
Q=0.0
0.3
0.4
0.5
0.7
0.9
1.0
1.2
1.4
2.0
5.0
10.0
Zero Current Switching Region
ZCS区域
ZVS区域1
Voltage Gain
Min. Operating Frequency
(Min. Vin, Max. Load)
Fs=Frp
Fs=Fr
Ideal Nominal Operating Frequency
(Nominal Vin, Load Independent)
Max. Operating Frequency
(Max. Vin , No load)
Fs/Fr

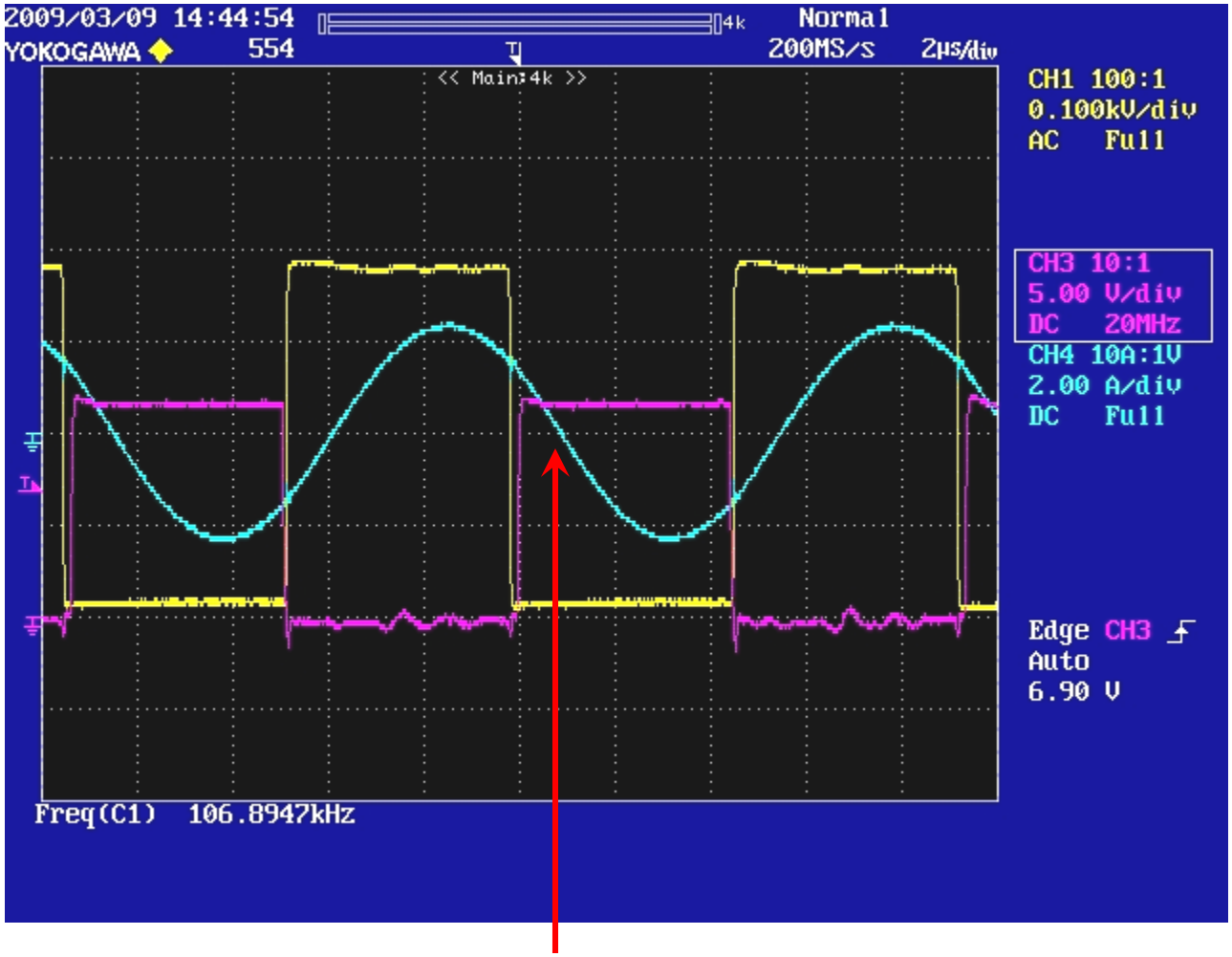
ZVS区域1的波形



f>fr 开关频率大于谐振频率

上管开通前电流由S-D流通

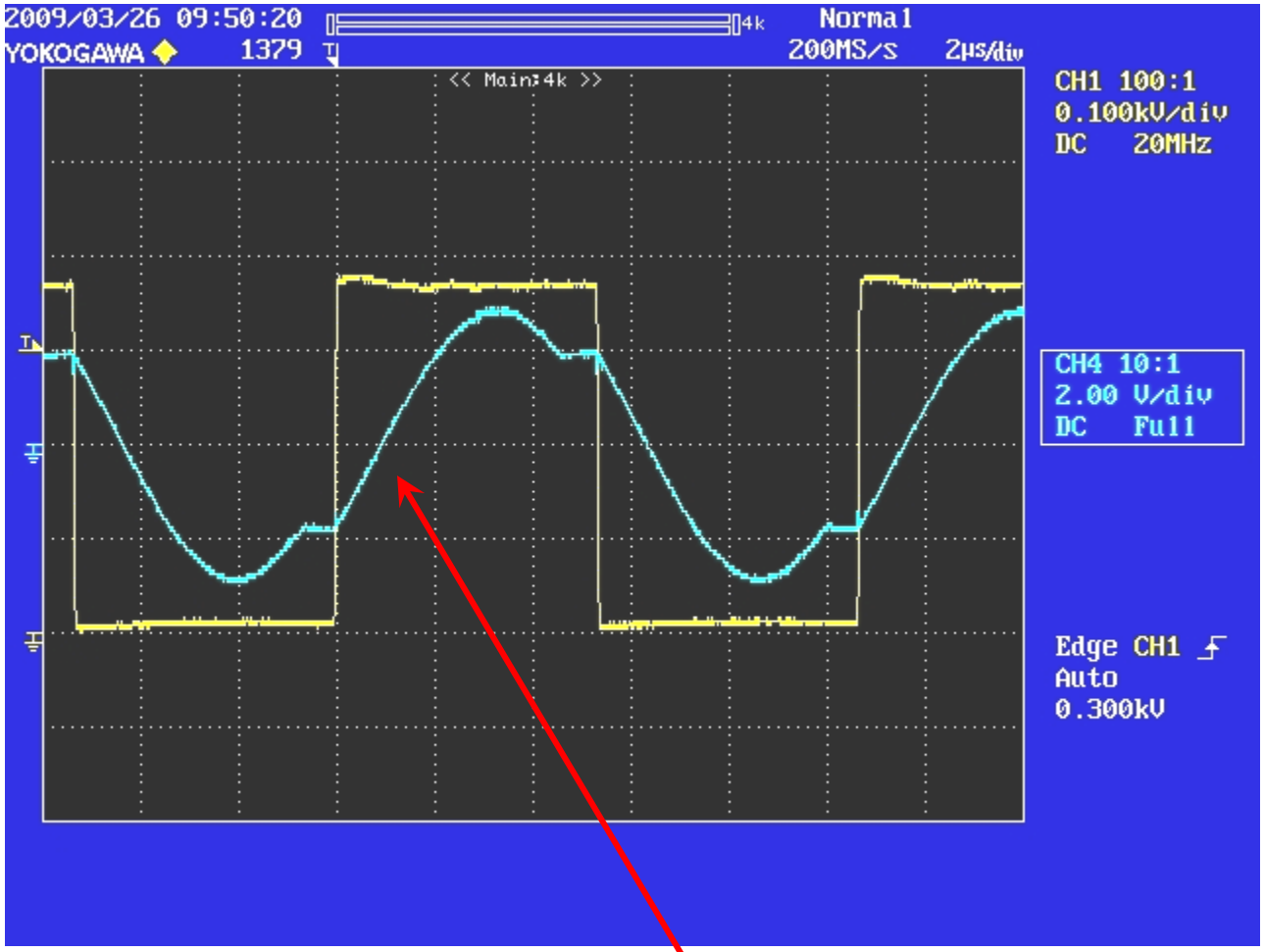
谐振点的波形



f=fr 开关频率等于谐振频率

下管开通前电流由S-D流通

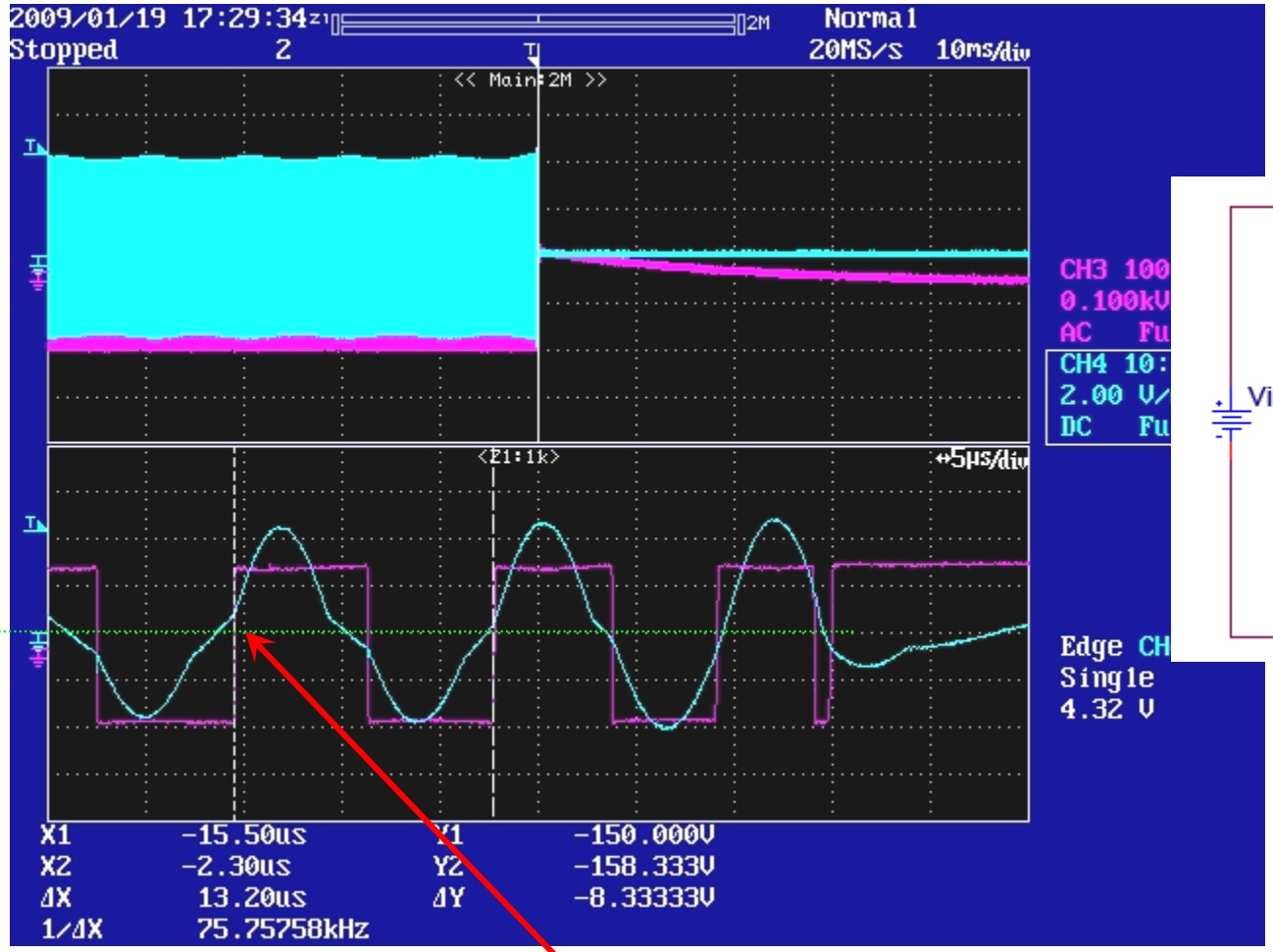
ZVS区域2的波形



f<fr 开关频率小于谐振频率

上管开通前电流由S-D流通

ZCS区域的波形



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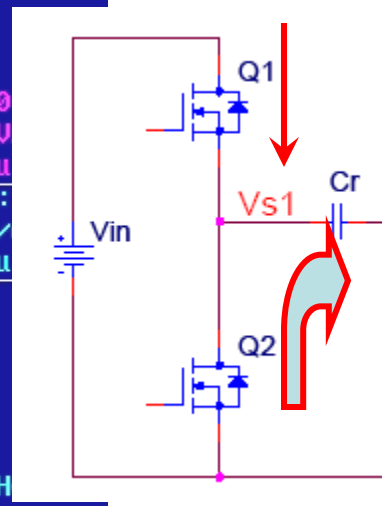


电流由D-S流过Q2体二极管

f<fr2 开关频率小于谐振频率

上管开通前电流由D-S流通

LLC的设计方法

已知的条件

1. 输入电压范围
2. 输出电压、电流
3. 确定需要的谐振频率
4. 额定输入、输出满载时电源工作在fr附近

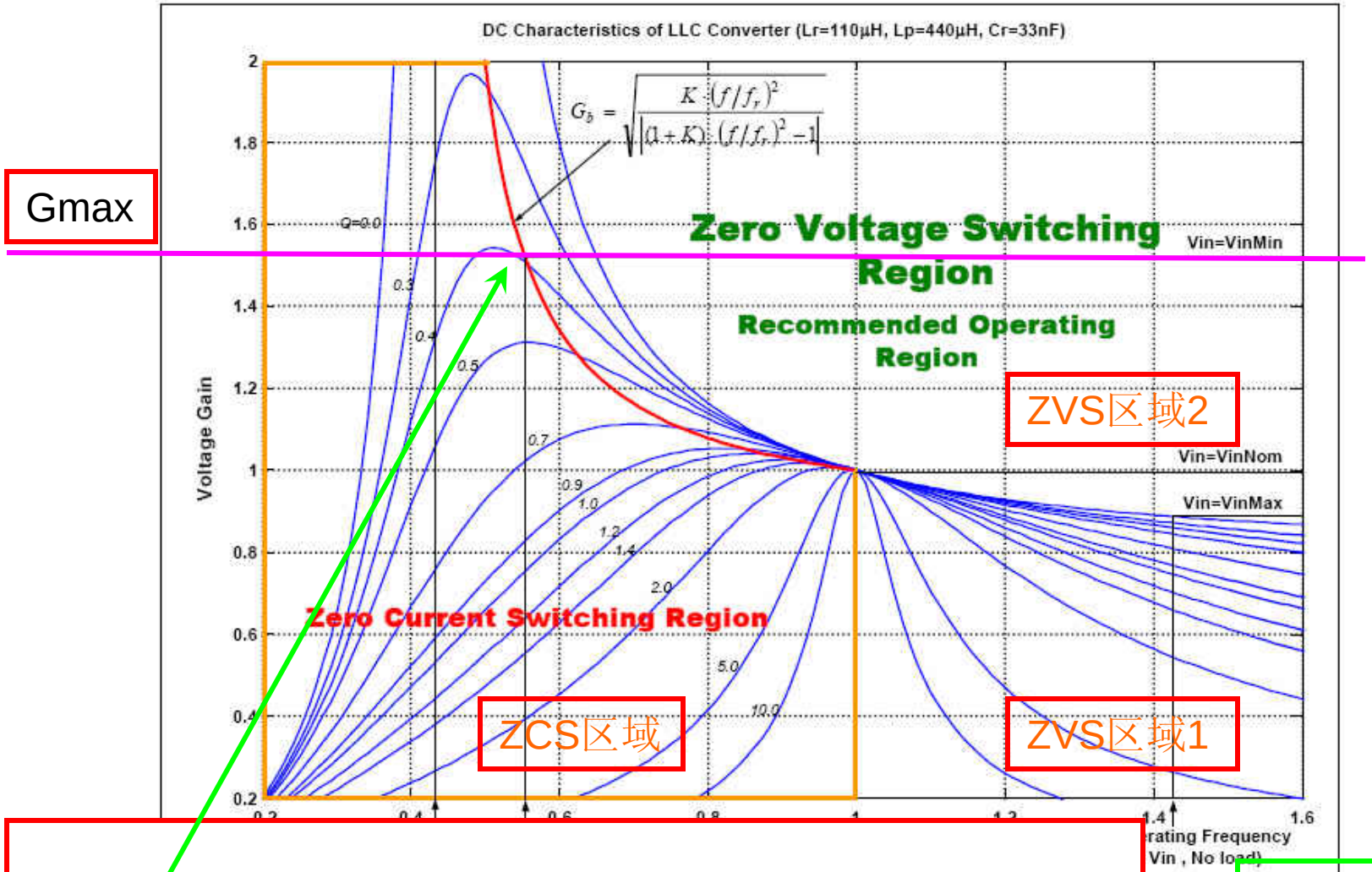
可简单得到的结果

1. 变压器变比n
2. 需要的电压增益Gmax,Gmin
3. Rac；k

需要求解的量

1. Q值，由此得到Cr,Ls,Lp
2. 最小、最大开关频率

LLC的DC特性

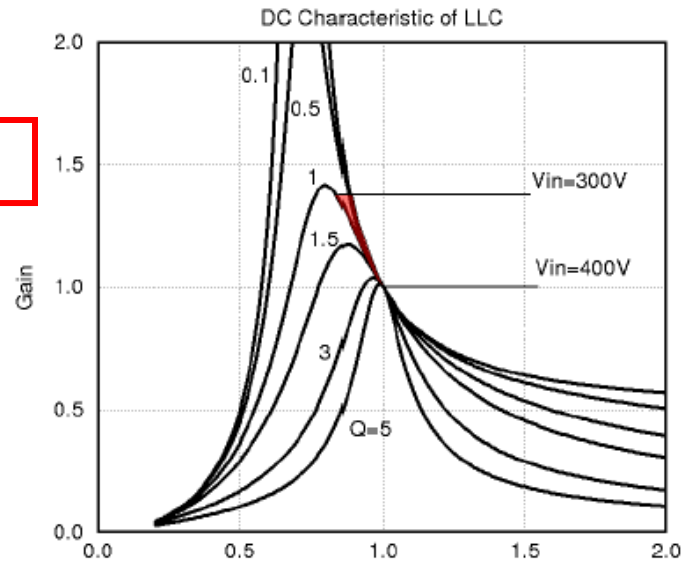


$$|G(f)| = \frac{k*(x)^2}{\sqrt{\{(1+k)*(x)^2-1\}^2+\{Qk(x)*[(x)^2-1]\}^2}}$$

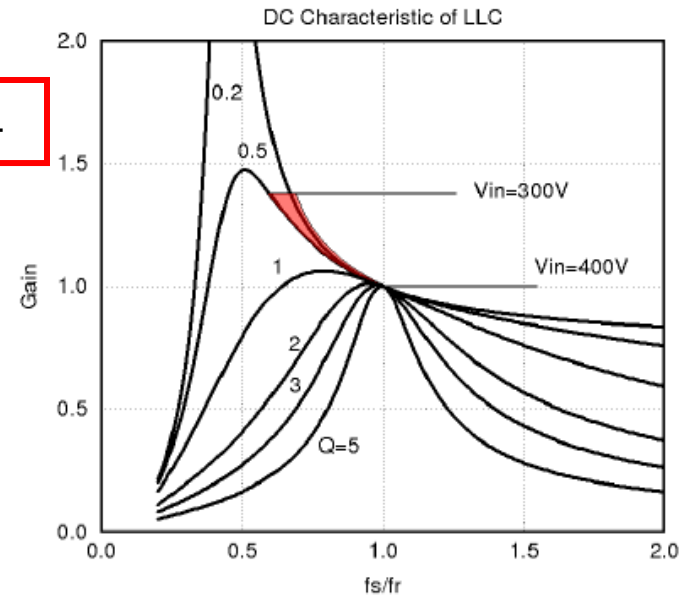
*Gmax*已知，3个变量，无法求出Q值

K值的确定

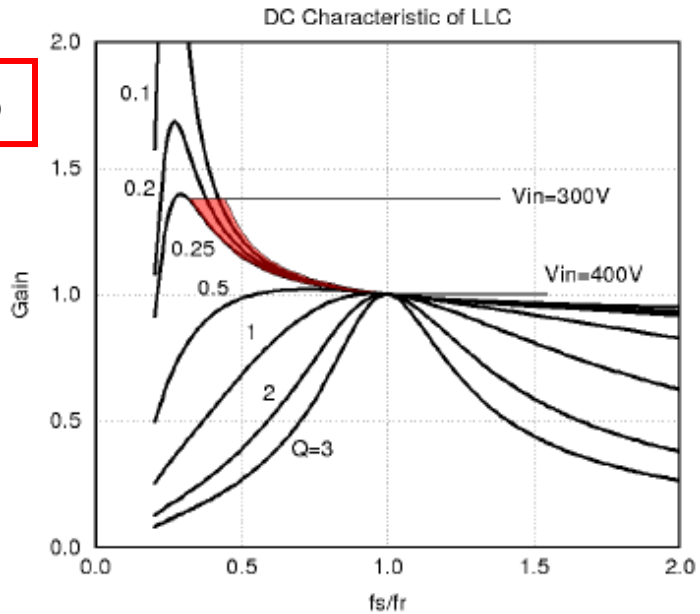
k=1



k=4



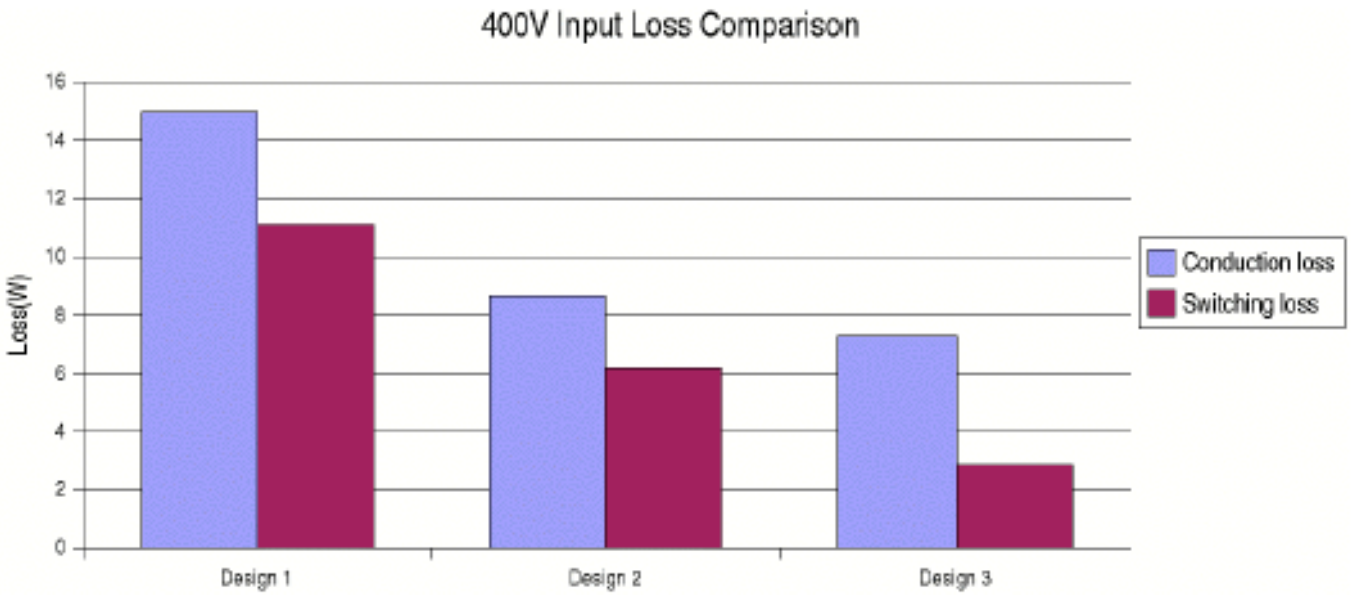
k=16



K值越小，获得相同增益的频率变化范围越窄

K 值越大，获得相同增益的频率变化范围越宽

K值的确定



k=1　　k=4　　k=16

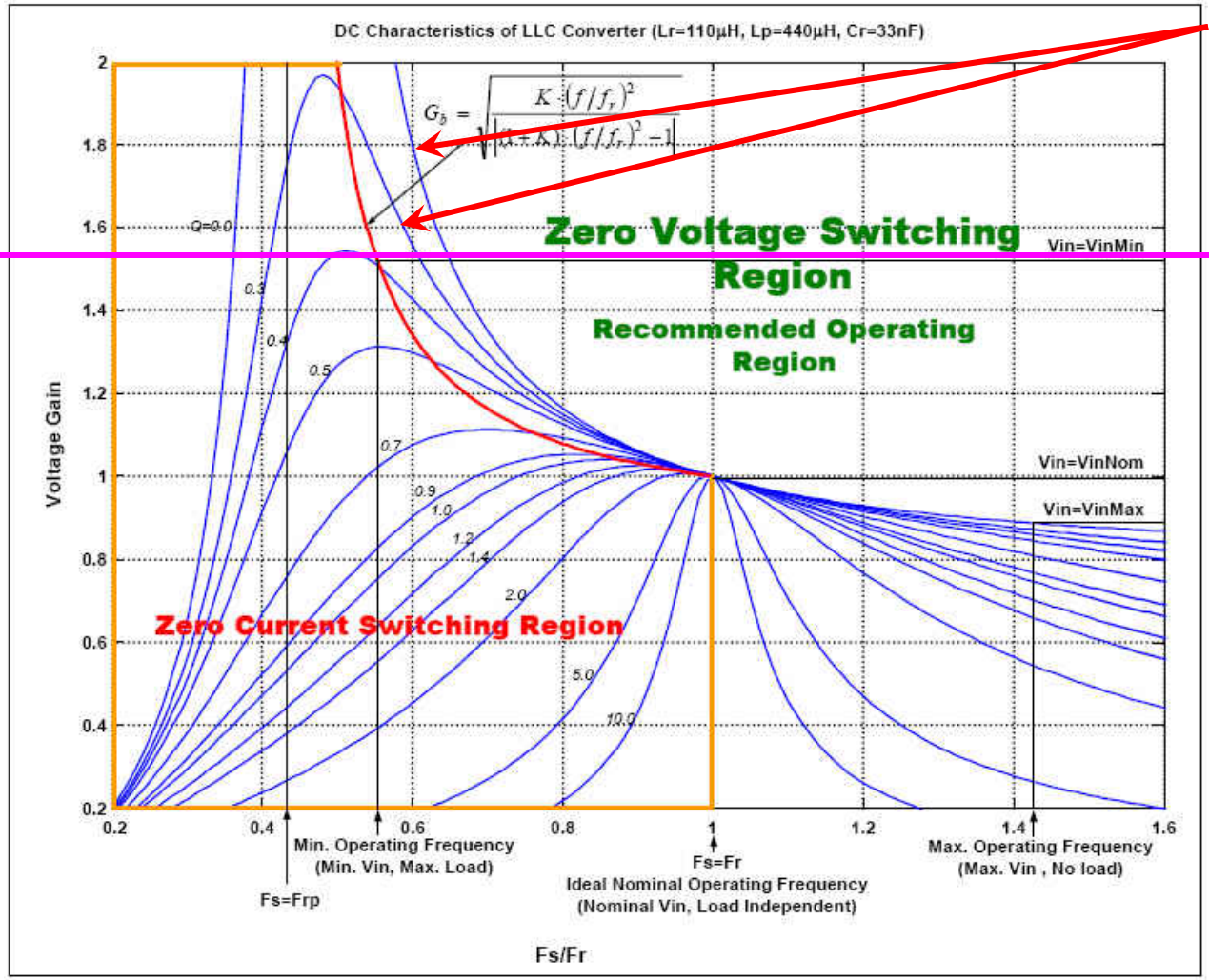
k值越大，MOSFET在fr附近的导通损耗和开关损耗越低

综合以上考虑k一般取2.5-6的范围

Q值的讨论

满足Gmax的Q值有很多

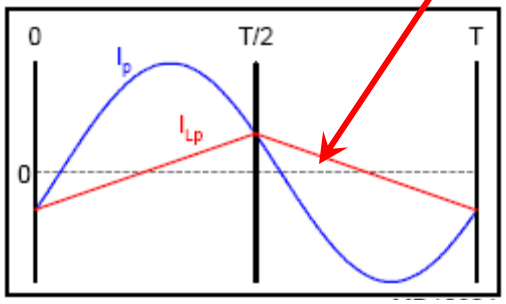
Gmax

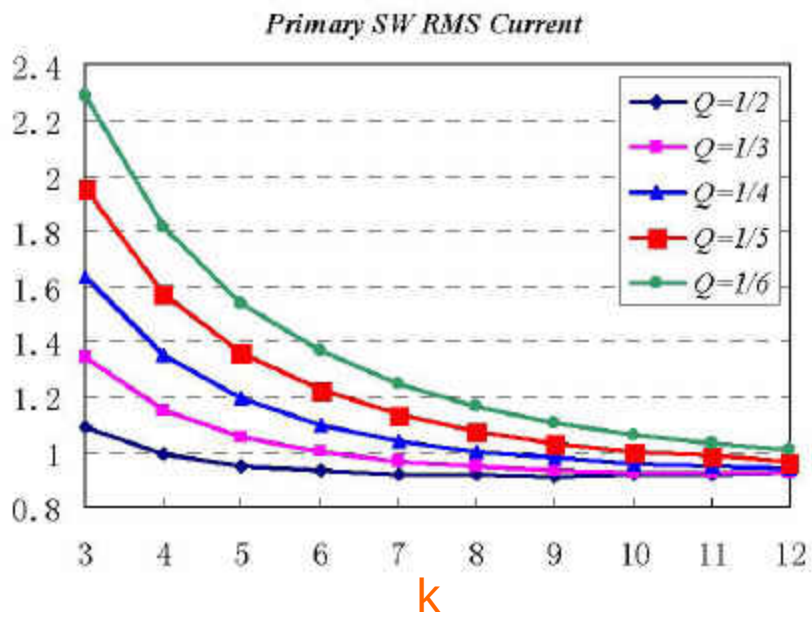


Q对初级电流的影响

$$Q=\frac{2\pi f_r L_s}{R_{AC}} \Rightarrow Q\downarrow \Rightarrow Ls\downarrow \Rightarrow Lp=k*Ls\downarrow \Rightarrow L_m=(Ls+Lp)\downarrow \Rightarrow I_{LP}\uparrow$$

$$I_{rms}=\frac{V_O}{8nR_L}\sqrt{\frac{2n^4R_L^2}{L_m^2 f_r^2}+8\pi^2}\uparrow \Rightarrow \eta\downarrow$$





K值固定后，在保证ZVS的条件下尽量选用大的Q值

再看变量

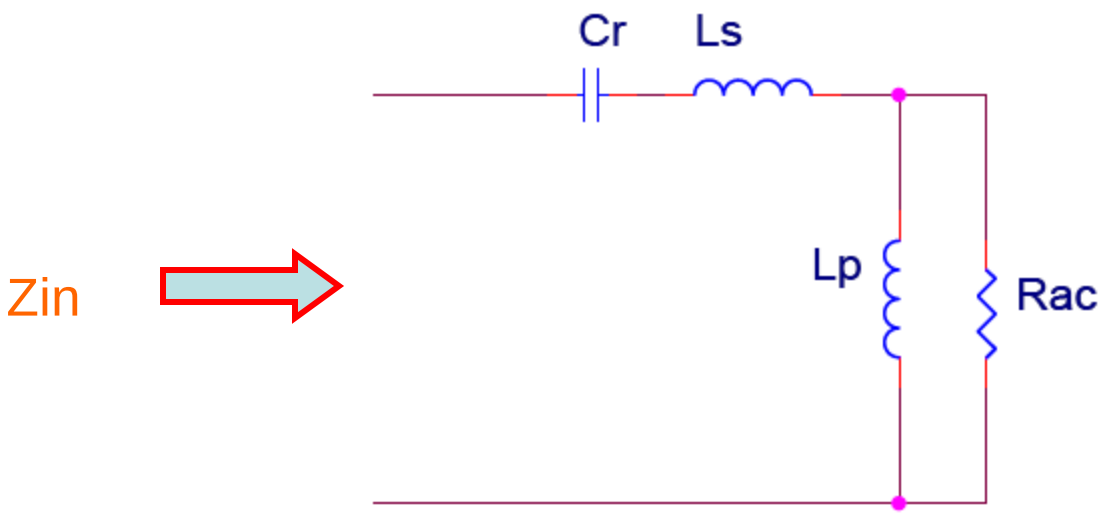
$$|G_{\max}| = \frac{k*(x)^2}{\sqrt{\{(1+k)*(x)^2-1\}^2+\{Qk(x)*[(x)^2-1]\}^2}}$$

已知量

仍有Q, x 为未知量，需要新的条件才能解出Q值，从而确定Cr, Lr, Lm

从阻抗想办法

LLC的阻抗特性



$$x = \frac{f}{fr}$$

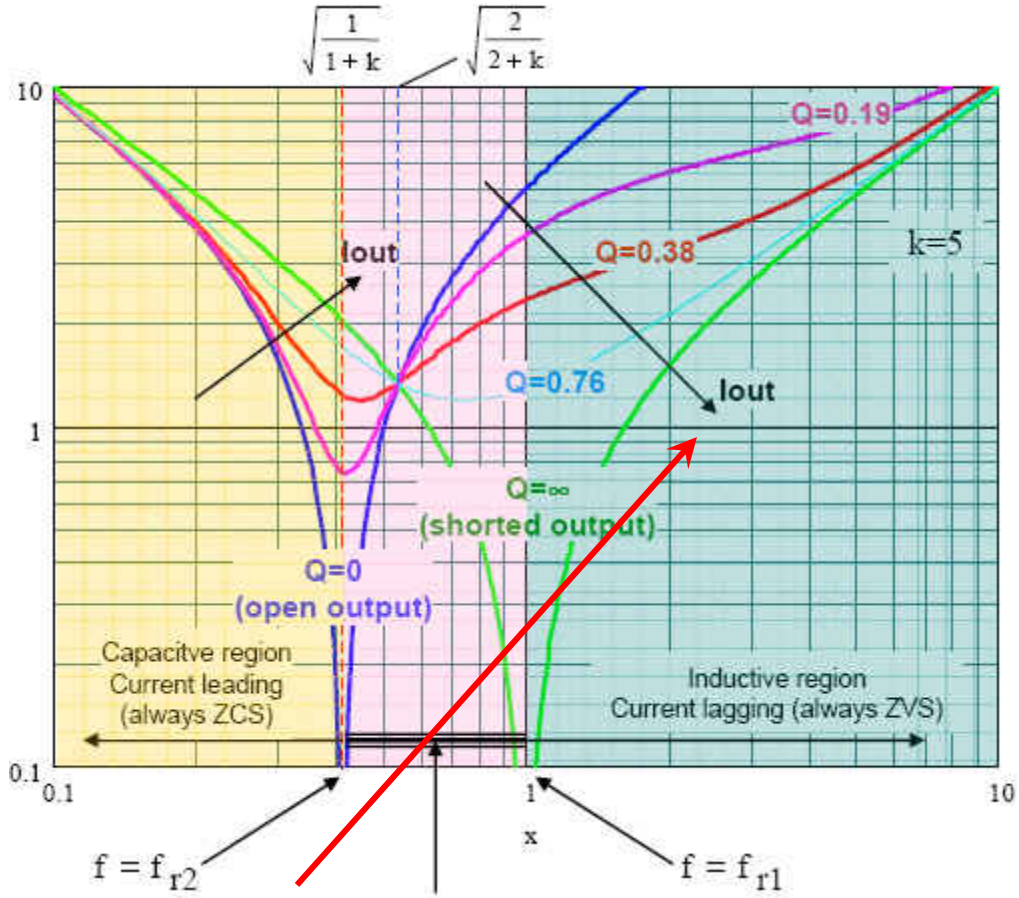
$$k = \frac{Lp}{Ls}$$

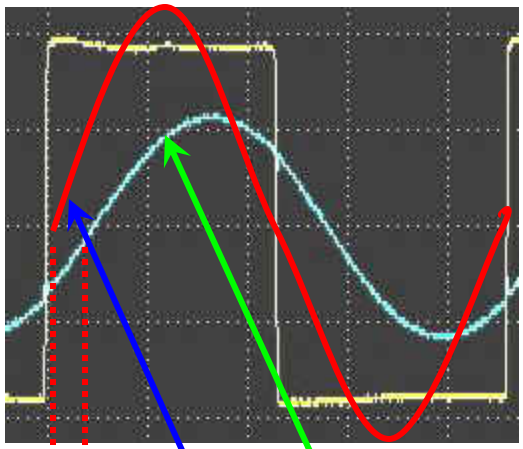
$$Q = \frac{2\pi f_r L_r}{R_{AC}}$$

$$Zin = \frac{1}{sCr} + sLr + (sLp // Rac)$$

$$= Zo * \{Q * \frac{x^2k^2}{1+x^2k^2Q^2} + j[x - \frac{1}{x} + \frac{xk}{1+x^2k^2Q^2}]\}$$

LLC的输入阻抗





Ø

电压波形

电流波形

ZVS的条件:电流落后于电压，即Zin为感性——频率增加，阻抗增加

相位差Ø为0是感性容性的分界线

LLC的阻抗特性

虚部为零，是感性容性的分界线，由此条件得到

$$Zin = Zo*\left(Q*\frac{x^2k^2}{1+x^2k^2Q^2}+j\left[x-\frac{1}{x}+\frac{xk}{1+x^2k^2Q^2}\right]\right)$$

$$x-\frac{1}{x}+\frac{xk}{1+x^2k^2Q^2}=0 \Rightarrow Q_{\max}(x)=\sqrt{\frac{1}{k(1-x^2)}-\frac{1}{(kx)^2}}$$

再看增益公式，把Qmax(x)带入公式，可求出x，再得到Qmax，从而得到Cr, Lr, Lm；令Q=0得空载工作频率

$$|G_{\max}|=\frac{k*(x)^2}{\sqrt{\{(1+k)*(x)^2-1\}^2+\{Q*k*(x)*[(x)^2-1]\}^2}}$$

已知量

代换为x

只有x是未知量

解求各参数

由最高输入电压时的增益(Gmin)和空载条件(Q=0)求解最高工作频率

$$|G_{\min}| = \frac{k*(x)^2}{\sqrt{\{(1+k)*(x)^2-1\}^2+\{Qk(x)*[(x)^2-1]\}^2}}$$

已知量

令Q=0

$$|G_{\min}| = \frac{k*(x)^2}{(1+k)*(x)^2-1}$$

只有x是未知量

结果

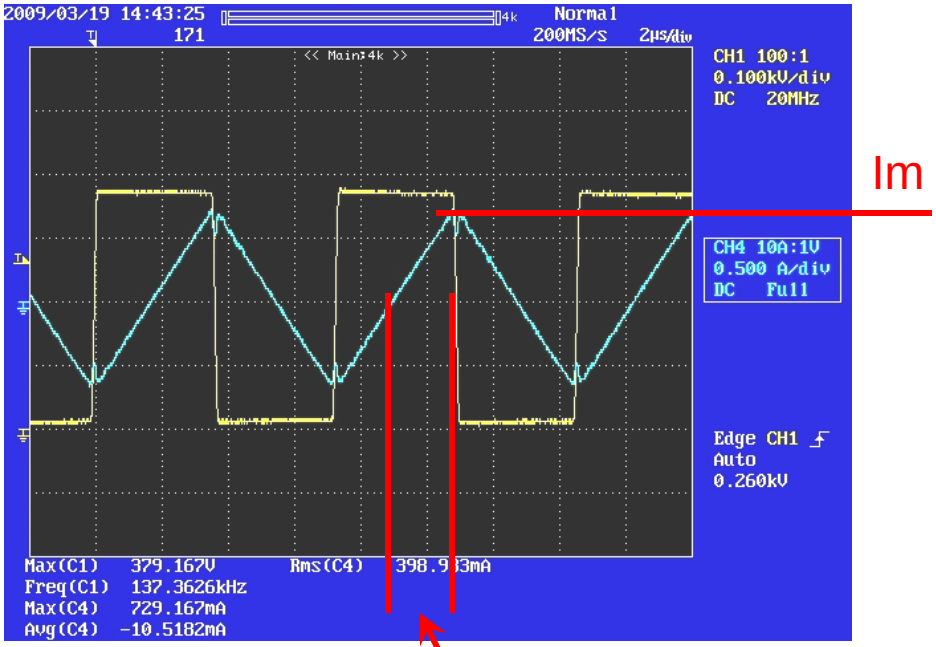
$$f_{\min} = \frac{f_r}{\sqrt{1 + k(1 - \frac{1}{G_{\max}^2})}} \qquad f_{\max} = \frac{f_r}{\sqrt{1 + k(1 - \frac{1}{G_{\min}})}}$$

$$Q = 0.95 Q_{\max} = \frac{0.95}{k * G_{\max}} * \sqrt{k + \frac{G_{\max}^2}{G_{\max}^2 - 1}}$$

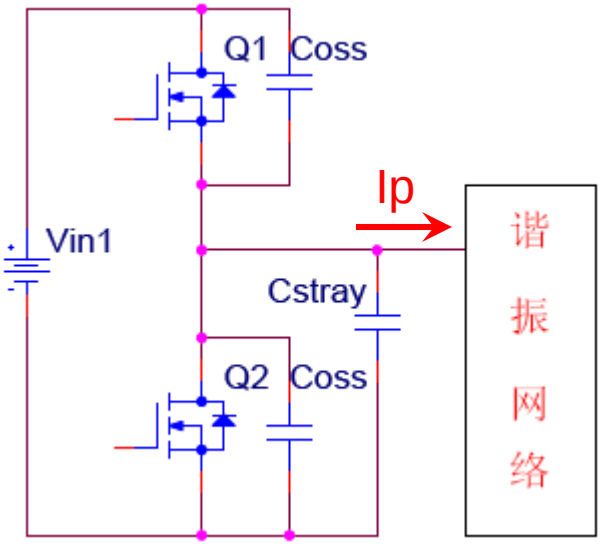
$$L_s = \frac{Q * R_{AC}}{2\pi f_r} \qquad C_r = \frac{1}{2\pi * f_r * R_{AC} * Q}$$

$$L_p = k * L_s$$

ZVS的另一个限制



Im



$$I_p > (2C_{oss} + C_{stray})\frac{V_{in}}{T_d}$$

保证ZVS的Ip

核算Im>Ip，
不满足重选Q

$$\mathrm{Im} = \frac{V_{inMax}}{L_s + L_p} * \frac{T_{\min}}{4}$$

$$= \frac{V_{inMax}}{4 f_{\max} * (L_s + L_p)}$$

输入电压最
高时的Im

归一化的计算步骤

1. 确定输入输出指标
2. 选择谐振频率和选择操作区域
3. 计算变压器变比和谐振元件值
4. 计算功率器件电压电流应力
5. 选择器件和变压器设计

实际计算步骤

1. 输入输出指标

输入电压范围： V_{inMin}=250V, V_{inMax}=420V

额定输入电压：V_{inNom}=400VDC

输出电压电流（最大值）：24V/10A, 12V/4A

输出功率：P_o=24×10＋12×4＝288W

2. 选择谐振频率和工作区域

谐振频率fr=100KHz

额定输入输出时电源工作在fr

3. 计算变压器变比和谐振元件值

3.1 理论变比

$$n=\frac{\frac{V_{inNom}}{2}}{V_o+V_D}=\frac{\frac{400}{2}}{24+0.7}=8.1$$

实际计算步骤

3.2 最高、最低输入电压的增益

$$G_{\min} = 2n * \frac{(Vo + Vd)}{V_{inMax}} = 2 * 8.1 * \frac{(24 + 0.7)}{420} = 0.952$$

$$G_{\max} = 2n * \frac{(Vo + Vd)}{V_{inMin}} = 2 * 8.1 * \frac{(24 + 0.7)}{250} = 1.6$$

3.3 计算等效为24V输出的负载电阻和反射电阻

$$R_L = \frac{V_O^2}{P_O} = \frac{24^2}{288} = 2\Omega$$

$$R_{AC} = n^2 \frac{8}{\pi^2} R_L = 8.1^2 * \frac{8}{\pi^2} * 2 = 106.5$$

实际计算步骤

3.4 取k＝3

3.5 计算Q,fmin,fmax,Ls,Lp,Cr

$$Q=\frac{0.95}{k*G_{\max}}*\sqrt{k+\frac{G_{\max}^{2}}{G_{\max}^{2}-1}}=\frac{0.95}{3*1.6}*\sqrt{3+\frac{1.6^{2}}{1.6^{2}-1}}=0.426$$

$$f_{\min}=\frac{f_r}{\sqrt{1+k(1-\frac{1}{G_{\max}^{2}})}}=\frac{100}{\sqrt{1+3(1-\frac{1}{1.6^{2}})}}=59.5KHz$$

$$f_{\max}=\frac{f_r}{\sqrt{1+k(1-\frac{1}{G_{\min}})}}=\frac{100}{\sqrt{1+3(1-\frac{1}{0.952})}}=108.5KHz$$

实际计算步骤

$$C_r = \frac{1}{2\pi * f_r * R_{AC} * Q} = \frac{1}{2\pi * 100 * 106.5 * 0.426} = 35nF$$

$$L_s = \frac{Q * R_{AC}}{2\pi f_r} = \frac{0.426 * 106.5}{2\pi * 100} = 72uH$$

$$L_p = k * L_r = 3 * 72 = 216uH$$

3.6 核算Im>Ip

$$\mathrm{I_m} = \frac{V_{inMax}}{4 f_{\max *}(L_r + L_p)} = \frac{420}{4 * 108.5 * (72 + 216)} = 3.36A$$

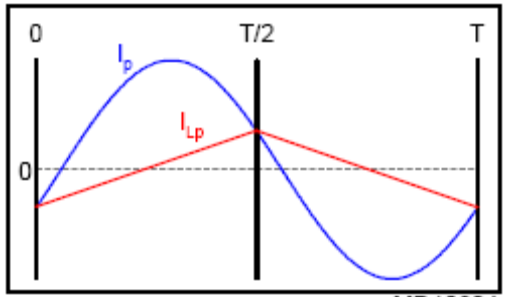
$$I_p = (2C_{oss} + C_{stray})\frac{V_{in}}{T_d} = 500 * 10^{-12} * \frac{420}{200 * 10^{-9}} = 1.05A$$

$$\mathrm{I_m} > I_p$$

如不满足需降低Q或增大Lr+Lp

实际计算步骤

4. 计算功率器件电压电流应力



4.1 初级电流有效值

$$I_{rms} = \frac{V_O}{8nR_L}\sqrt{\frac{2n^4R_L^2}{L_m^2 f_r^2} + 8\pi^2} = 1.6A$$

4.2 MOSFET电压,电流最大值,电流有效值

$$V_{Mos} = V_{inMax} = 420V$$

$$I_{Max_Mos} = I_{OCP}$$

$$I_{rms_Mos} = \frac{I_{rms}}{\sqrt{2}} = \frac{1.6}{\sqrt{2}} = 1.13A$$

$$P_{Conduct_loss} = I_{rms_Mos}^2 R_{ds} = 1.28R_{ds}$$

实际计算步骤

4.3 次级整流管电压,电流,损耗(24V输出）

$$V_{D_Max} = 2*V_O = 2*24 = 48$$

$$I_{D_Avg} = \frac{I_o}{2} = \frac{10}{2} = 5A(24V)$$

$$P_{D_Conduct_loss} = V_{D_Conduct_Avg} * I_{D_Avg} = 0.7*5 = 3.5W$$

4.4 谐振电容电流有效值、最大电压

$$I_{Cr_rms} = I_{rms} = \frac{V_O}{8nR_L}\sqrt{\frac{2n^4 R_L^2}{L_m^2 f_r^2} + 8\pi^2} = 1.6A$$

$$V_{Cr_Max} \cong \frac{V_{in_Max}}{2} + \sqrt{2} * I_{rms_Max} * \frac{1}{2\pi f_r C_r} = \frac{420}{2} + \frac{I_{OCP}}{2\pi f_r C_r}$$

实际计算步骤

4.5 输出电容的电流有效值(f=fr,24V输出)

$$I_{Co_Rms} = \sqrt{\left(\frac{\pi I_o}{2\sqrt{2}}\right)^2 - I_o^{\,2}} = \sqrt{\frac{\pi^2 - 8}{8} I_o} = 2.32A$$

实际计算步骤

5. 选择器件和变压器设计

MOSFET: 满足20％裕量,电压500V，电流从发热和Coss考虑(保证高压时ZVS)

Cr: 满足RMS电流的要求，电压为计算值1.5倍左右

Co:满足RMS电流要求

D: 电压满足20％裕量;电流考虑到不平衡,取40％裕量，其余从发热考虑

变压器实际变比

$$n_{real} = n * \sqrt{\frac{Lr + Lp}{Lp}} = n * \sqrt{\frac{k+1}{k}} = 8.1 * \sqrt{\frac{3+1}{3}} = 9.35$$

实际计算步骤

初级最小匝数(EER40)

$$N_{P_Min} = \frac{n_{real}(V_o + V_d)}{2 f_{\min} * \Delta B * A_e} = \frac{9.35(24+0.7)}{2*59.5*0.4*149} * 10^3 = 32.5$$

选择次级匝数，计算初级匝数使其大于32.5T

N_{12}=2T; N_{24}=4T

Np=9.35*4=37.4>33

最终结果：

Np=37T

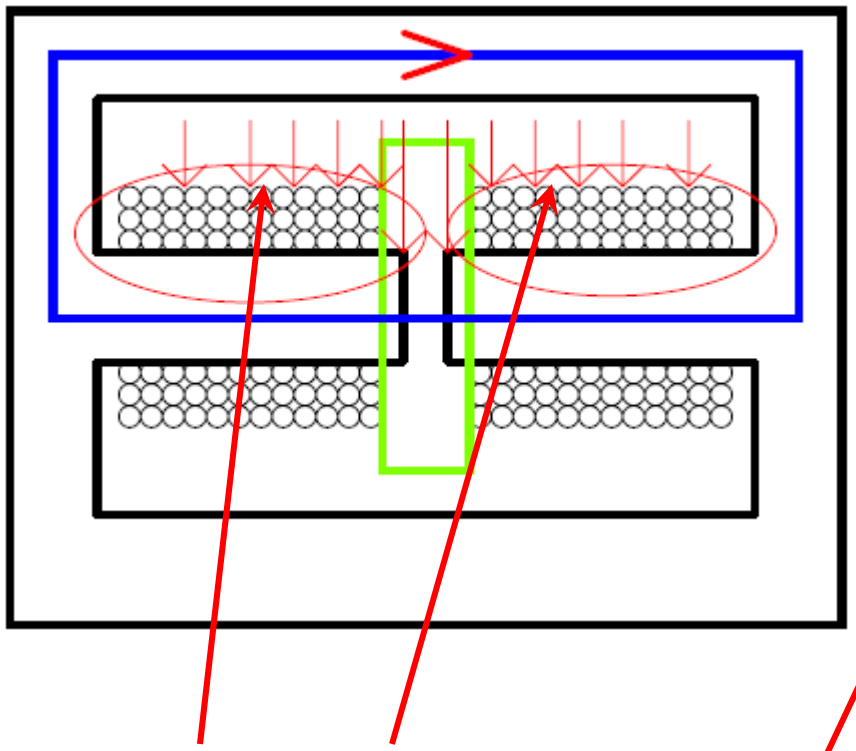
N_{12}=2T

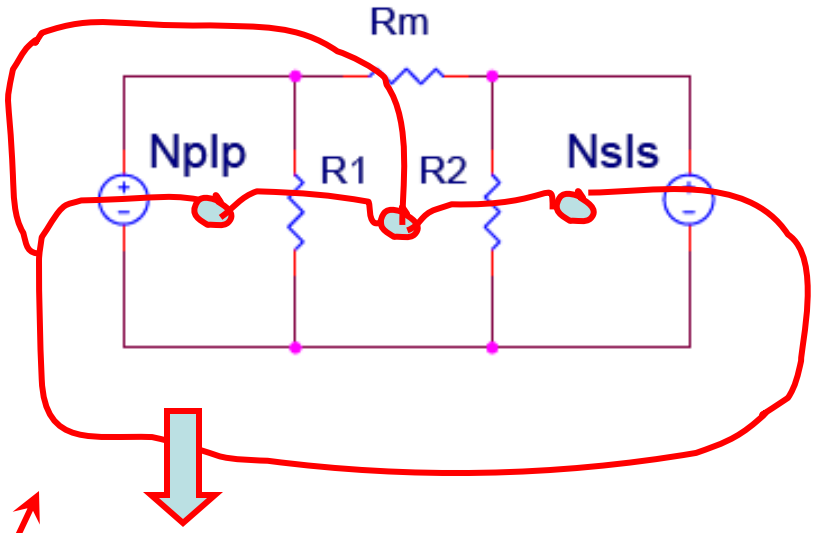
N_{24}=4T

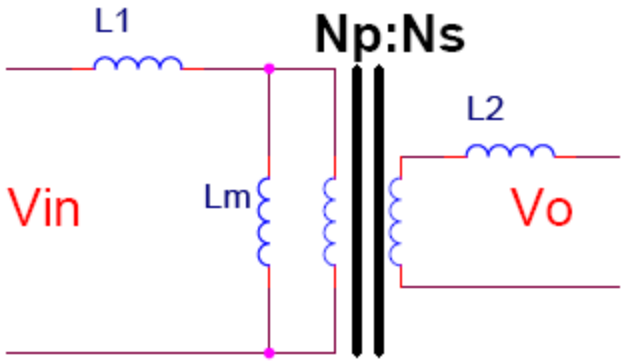
LLC的几个问题讨论

1. 变压器变比和结构
2. 电流不对称
3. 控制环路

LLC变压器磁阻模型



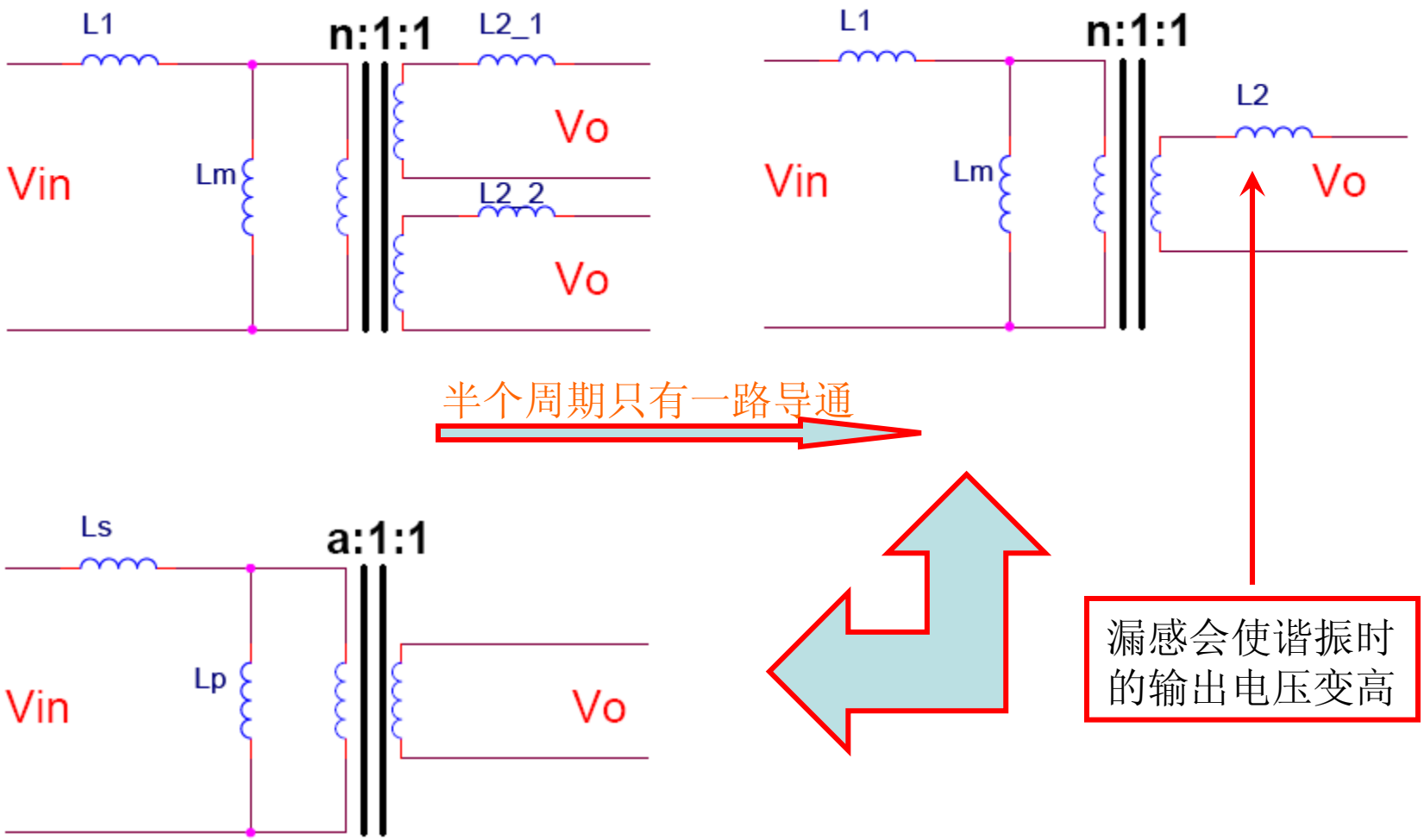




除了普通变压器的空气漏磁外,LLC变压器有新的漏磁路径，使漏感很大

磁电对偶关系得到电路模型

LLC变压器电路模型


L1
n:1:1
L2_1
Vo
Vin
Lm
L2_2
Vo
L1
n:1:1
L2
Vin
Lm
Vo
半个周期只有一路导通
Ls
a:1:1
Vin
Lp
Vo
漏感会使谐振时的输出电压变高

LLC变压器电路模型

图1

图2.

$$\begin{cases} V_{in} = L_1\dfrac{di}{dt} + L_m\dfrac{di_m}{dt} & ① \\ V_{in} = L_1\dfrac{di}{dt} + n\left(V_0 + L_2\dfrac{di_0}{dt}\right) & ② \\ i = i_m + i_0 = i_m + \dfrac{i_0}{n} & ③ \\ L_1 = n^2 L_2 & ④ \end{cases}$$

a为理论变比,n为实际变比

由③得 $i_m = i - \dfrac{i_0}{n}$. ⑤

把⑤代入①得

$$V_{in} = L_1\frac{di}{dt} + L_m\frac{d\left(i-\frac{i_0}{n}\right)}{dt} = L_1\frac{di}{dt} + L_m\frac{di}{dt} - \frac{L_m}{n}\cdot\frac{di_0}{dt} \quad ⑥$$

$$= (L_1+L_m)\frac{di}{dt} - \frac{L_m}{n}\cdot\frac{di_0}{dt} \quad ⑦$$

把④代入②得

$$V_{in} = L_1\frac{di}{dt} + nV_0 + n\cdot\frac{L_1}{n^2}\frac{di_0}{dt} = L_1\frac{di}{dt} + nV_0 + \frac{L_1}{n}\frac{di_0}{dt} \quad ⑦$$

⑥×L_1得

$$L_1\cdot V_{in} = L_1(L_1+L_m)\frac{di}{dt} - \frac{L_1\cdot L_m}{n}\frac{di_0}{dt} \quad ⑧$$

①×L_m得

$$\begin{cases} L_m\cdot V_{in} = L_1\cdot L_m\dfrac{di}{dt} + nL_mV_0 + \dfrac{L_1\cdot L_m}{n}\cdot\dfrac{di_0}{dt} & ⑨ \\ L_1V_{in} = L_1(L_1+L_m)\dfrac{di}{dt} - \dfrac{L_1\cdot L_m}{n}\cdot\dfrac{di_0}{dt} & ⑧ \end{cases}$$

⑧+⑨得

$$(L_1+L_m)V_{in} = L_1(L_1+2L_m)\frac{di}{dt} + nL_mV_0$$

$$⑮\quad V_{in} = \frac{L_1(L_1+2L_m)}{L_1+L_m}\cdot\frac{di}{dt} + \frac{nL_m}{L_1+L_m}\cdot V_0$$

$$\begin{cases} L_s = L_1 + L_m // n^2L_2 = L_1 + L_m // L_1 = L_1 + \dfrac{L_1\cdot L_m}{L_1+L_m} = \dfrac{L_1(L_1+2L_m)}{L_1+L_m} & ⑩ \\ L_s + L_p = L_1 + L_m = L. & ⑪ \end{cases}$$

由⑪得 $L_m = L - L_M$ ⑫

把⑫代入⑩得

$$L_s = L_1 + \frac{L_1\cdot L_m}{L_1+L_m} = L - L_m + \frac{(L-L_m)\cdot L_m}{L-L_m+L_m} = L - L_m + \frac{(L-L_m)L_m}{L}$$

两边同乘以L得

$$⑬\begin{cases} L\cdot L_s = L^2 - L\cdot L_m + L\cdot L_m - L_m^2 \Rightarrow L_m^2 = L^2 - L\cdot L_s \Rightarrow L_m = \sqrt{L^2 - L\cdot L_s} \\ L_1 = L - L_m = L - \sqrt{L^2 - L\cdot L_s} \end{cases}$$

把⑬代入⑮得

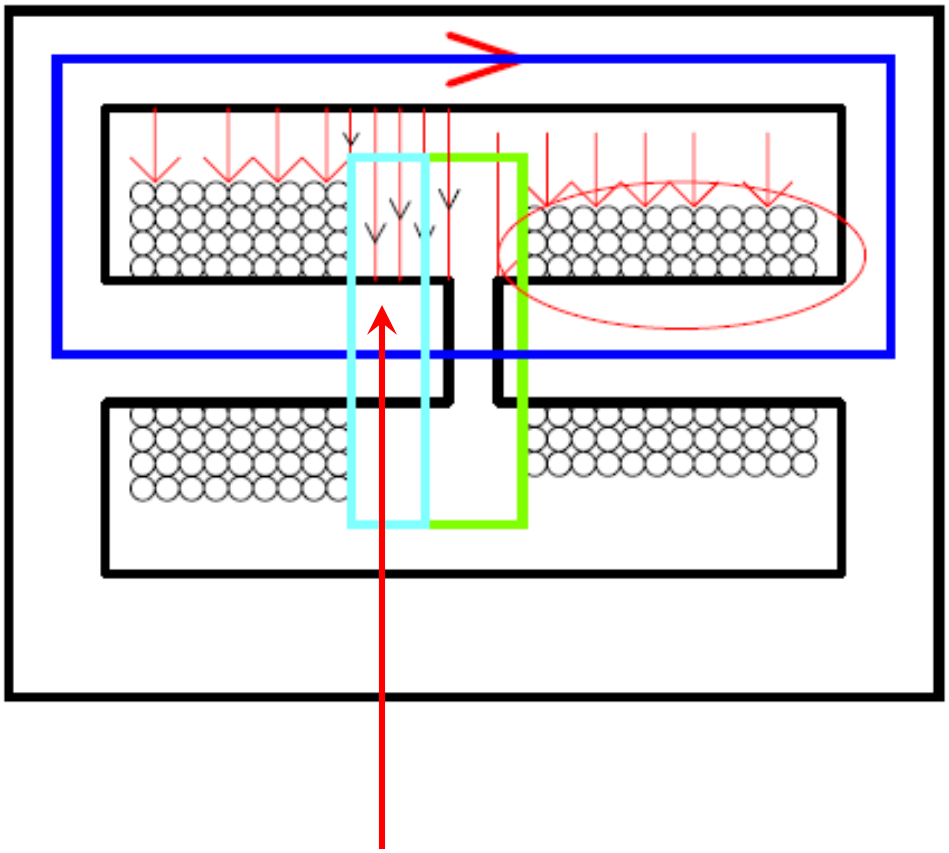
$$V_{in} = \left[L_1 + \frac{L_1\cdot L_m}{L_1+L_m}\right]\frac{di}{dt} + \frac{n\sqrt{L^2-L\cdot L_s}}{L}V_0 = L_s\frac{di}{dt} + n\sqrt{\frac{L-L_s}{L}}\cdot V_0$$

$$= L_s\frac{di}{dt} + n\sqrt{\frac{L_p}{L_s+L_p}}\cdot V_0$$

由图②得 $a = n\sqrt{\dfrac{L_s}{L_s+L_p}} \Rightarrow n = a\sqrt{\dfrac{L_s+L_p}{L_p}}$

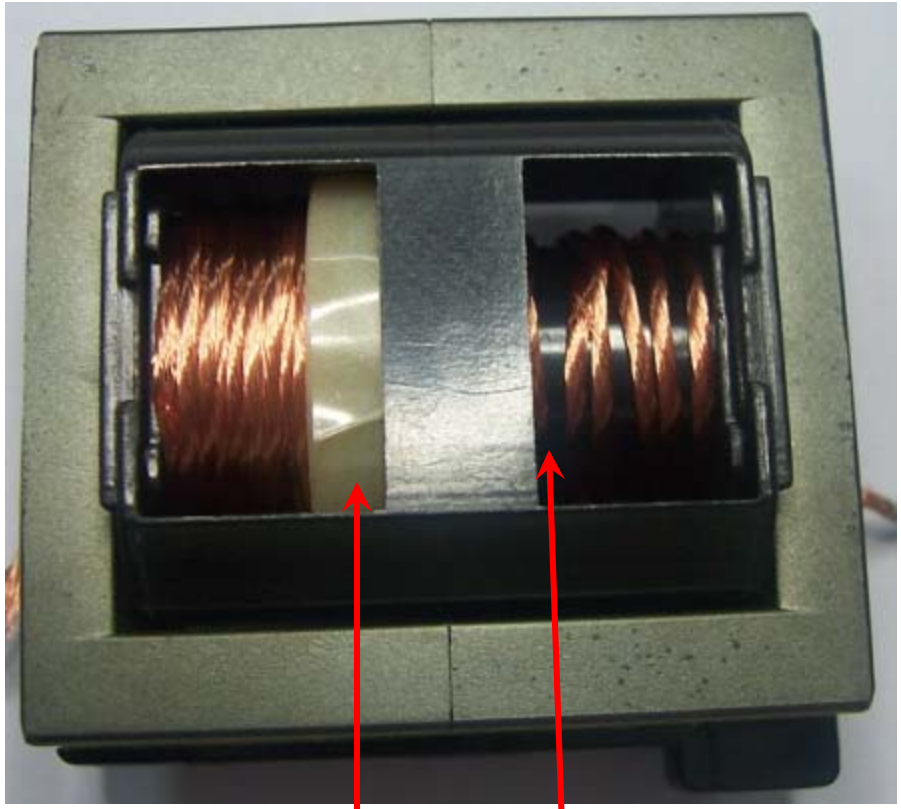
由于fr时的增益>1,实际变比比理论变比大才能得到理论电压

LLC变压器漏感的调整



增加初次级的距离增加了漏感

一个变压器实测结果



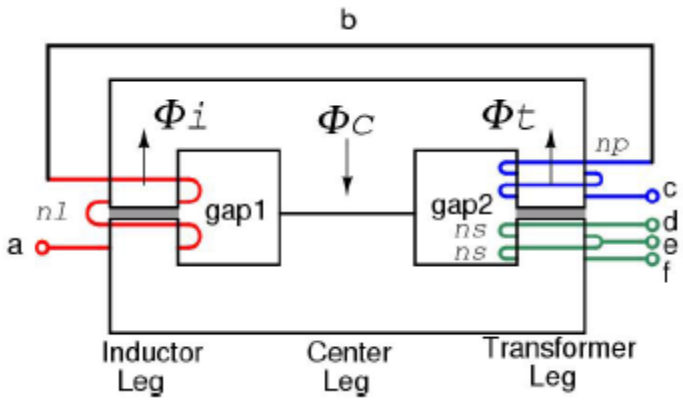
初次级都不加3.6mm档墙
Lm=680uH,Ls=123uH
在次级加3.6mm档墙
Lm=680uH,Ls=140uH

初级加3.6mm,次级不加3.6mm档墙
Lm=700uH,Ls=146uH
初级、次级都加3.6mm档墙
Lm=700uH,Ls=160uH

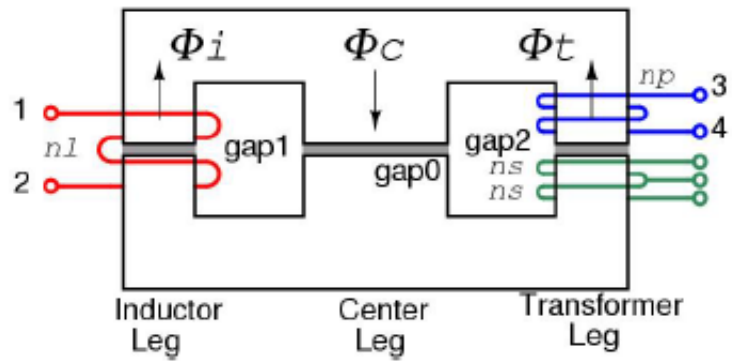
档墙

次级加的位置

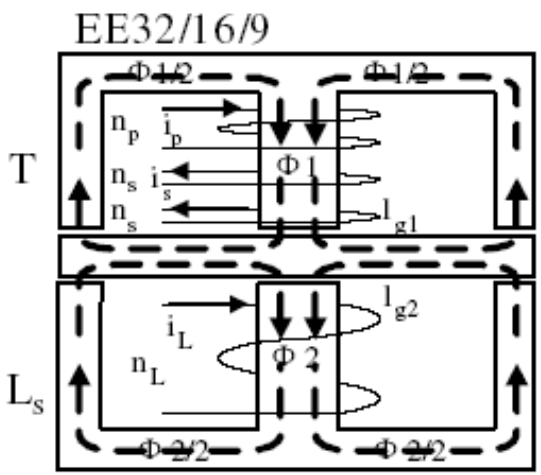
可能的变压器集成方式



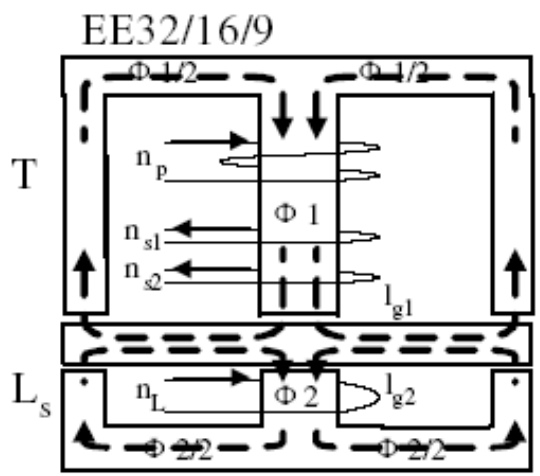
A



B

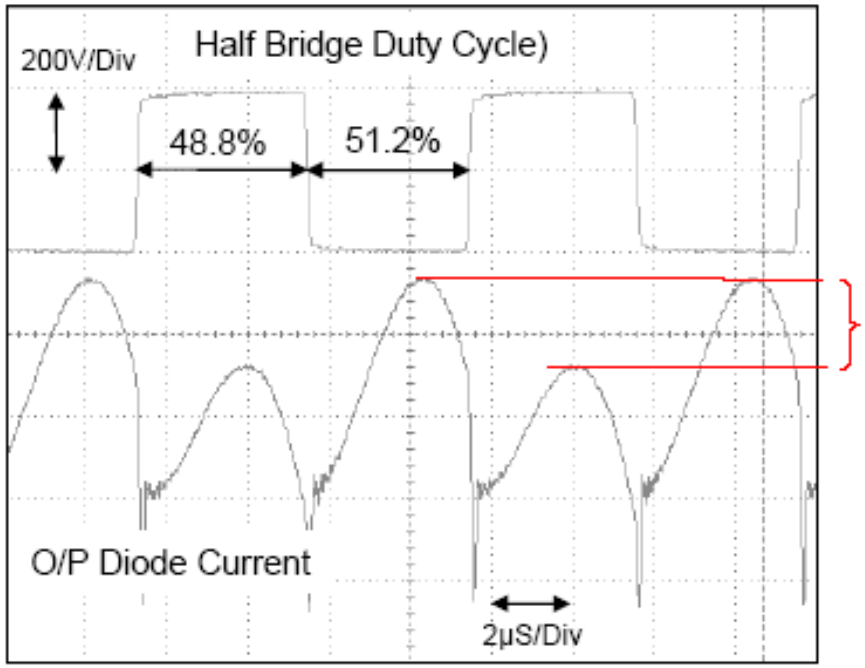


C



D

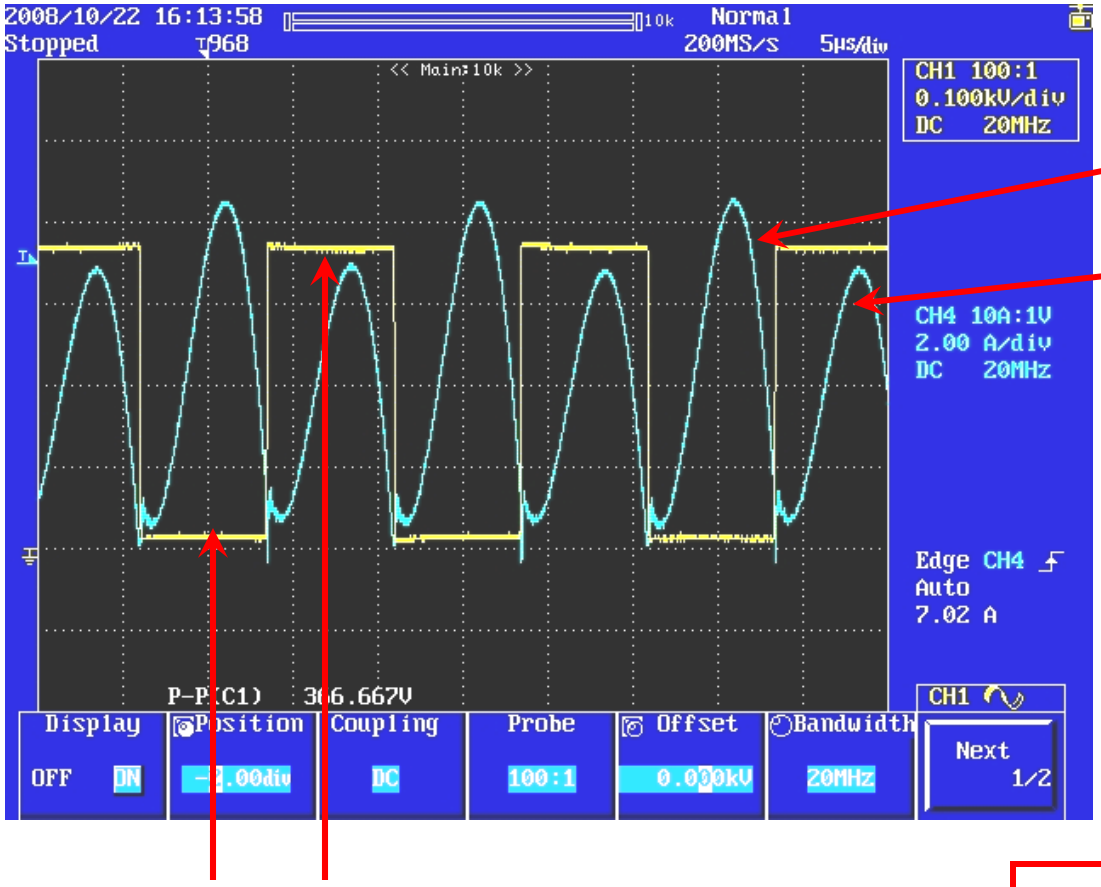
驱动不对称造成电流不对称



实际测量结果：

- 二极管经受2.4%的占空比失衡时，就会存在±20%的RMS电流失配
- 导致二极管过度设计

驱动对称就好了吗？



电流不平衡由正负半周的漏感不同引起

驱动几乎完全对称

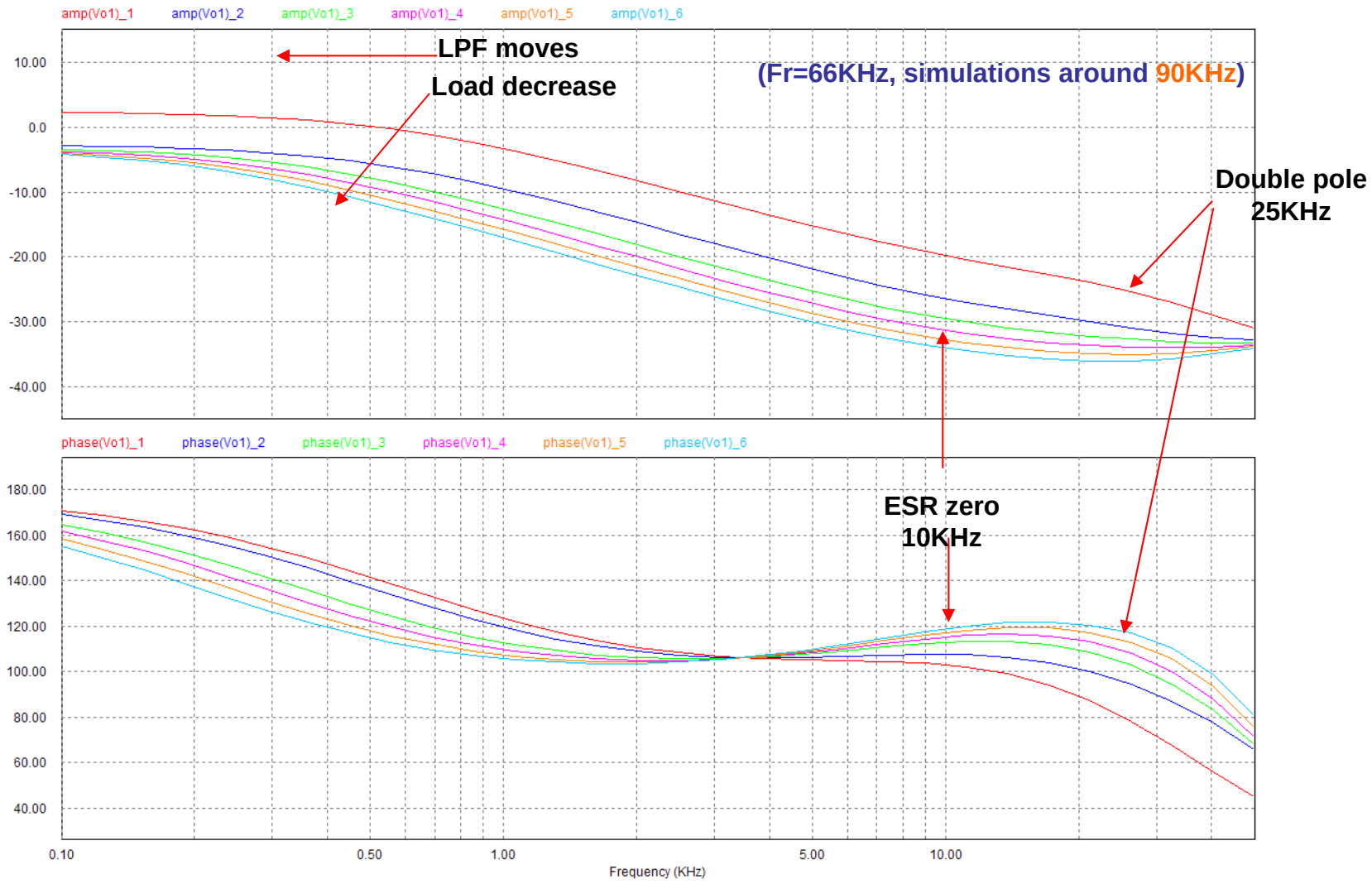
双线并绕可以减少这种情况

LLC控制环路

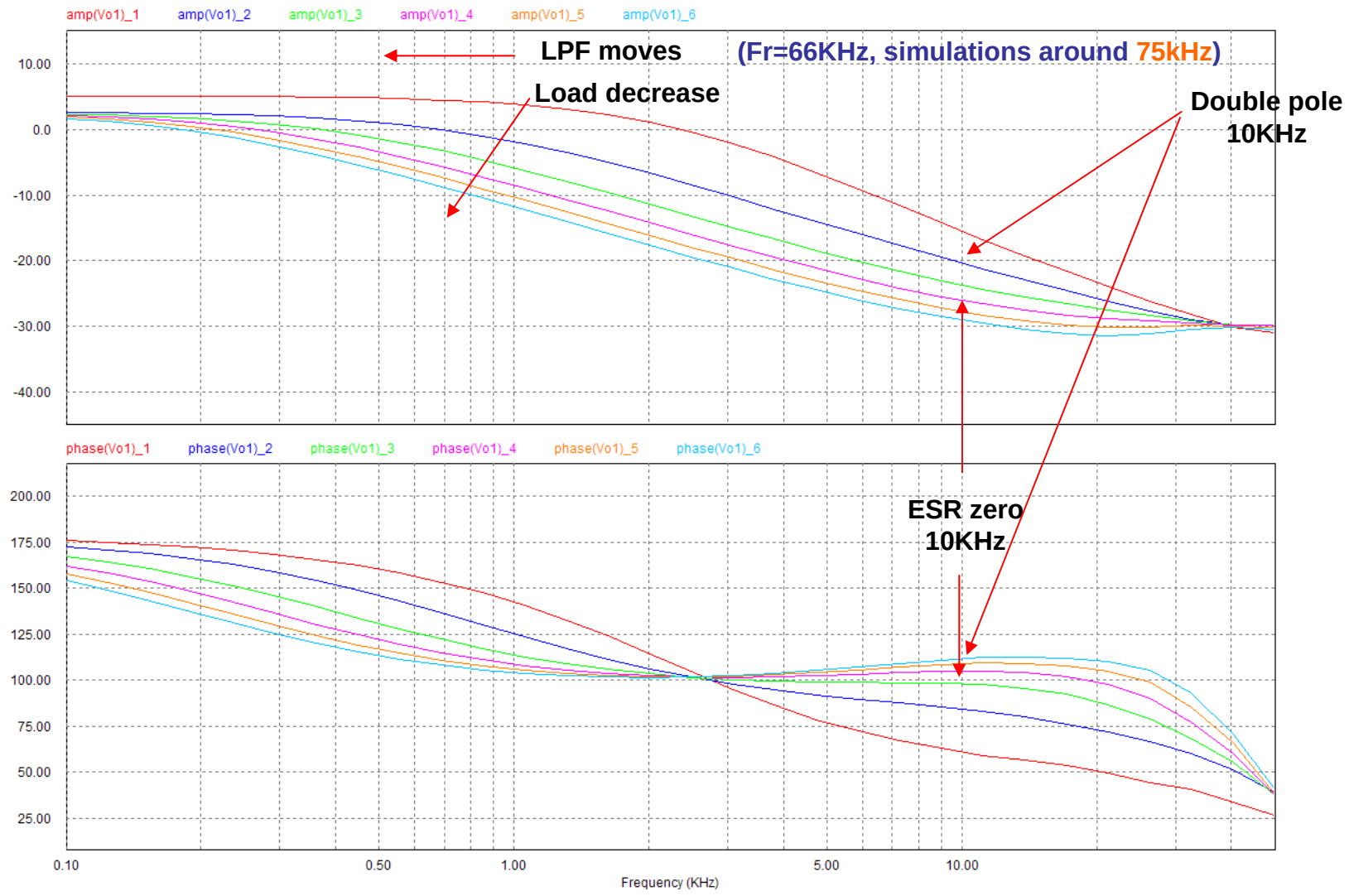
1. 由于调制频率在谐振频率范围，LLC不适合于状态空间法
2. 见于论文的是扩展描述法，但相当复杂
3. 目前常用的是时域仿真的方法

优点：只需要开关模型，很多软件工具可用，如Simplis。
结果和实际一致（其实就相当于网络分析仪测试）
缺点：无法取得零、极点的数学表达，所以不能用数学
工具设计反馈

高于谐振频率不同负载时的小信号传递特性1



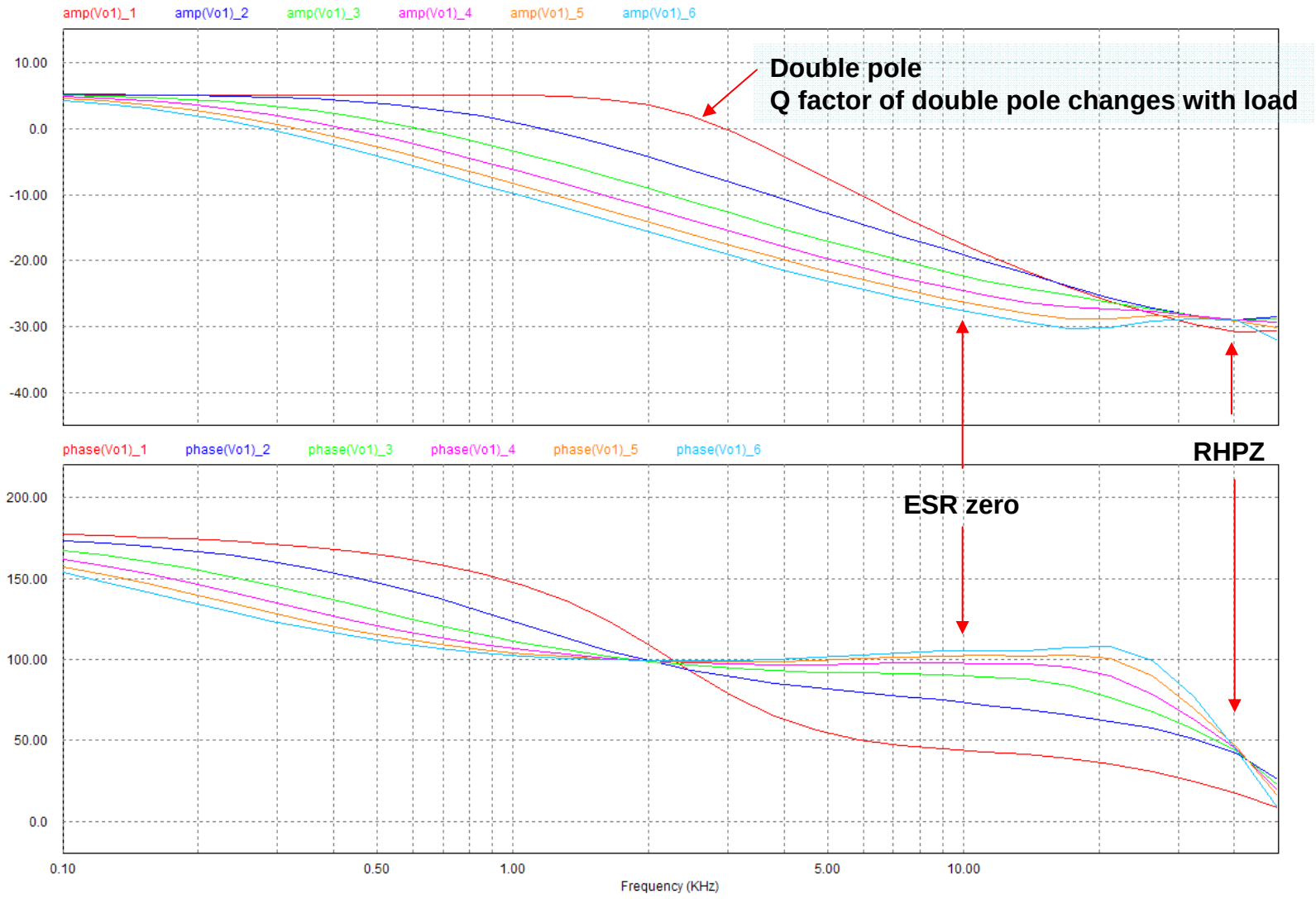
高于谐振频率不同负载时的小信号传递特性2



高于谐振频率不同负载时的小信号传递特性

1. 一个零点，有电容的ESR形成
2. 一个低频极点，其位置与输出电容、负载和谐振电路参数有关当负载减小时极点的位置往低频移动
3. 差频双极点,频率在(f-fr)，当开关频率接近谐振频率时，双极点分成两个极点(低Q值双极点)，一个接近于低频极点，一个在比较高的频率

谐振频率处不同负载时的小信号传递特性



谐振频率处不同负载时的小信号传递特性

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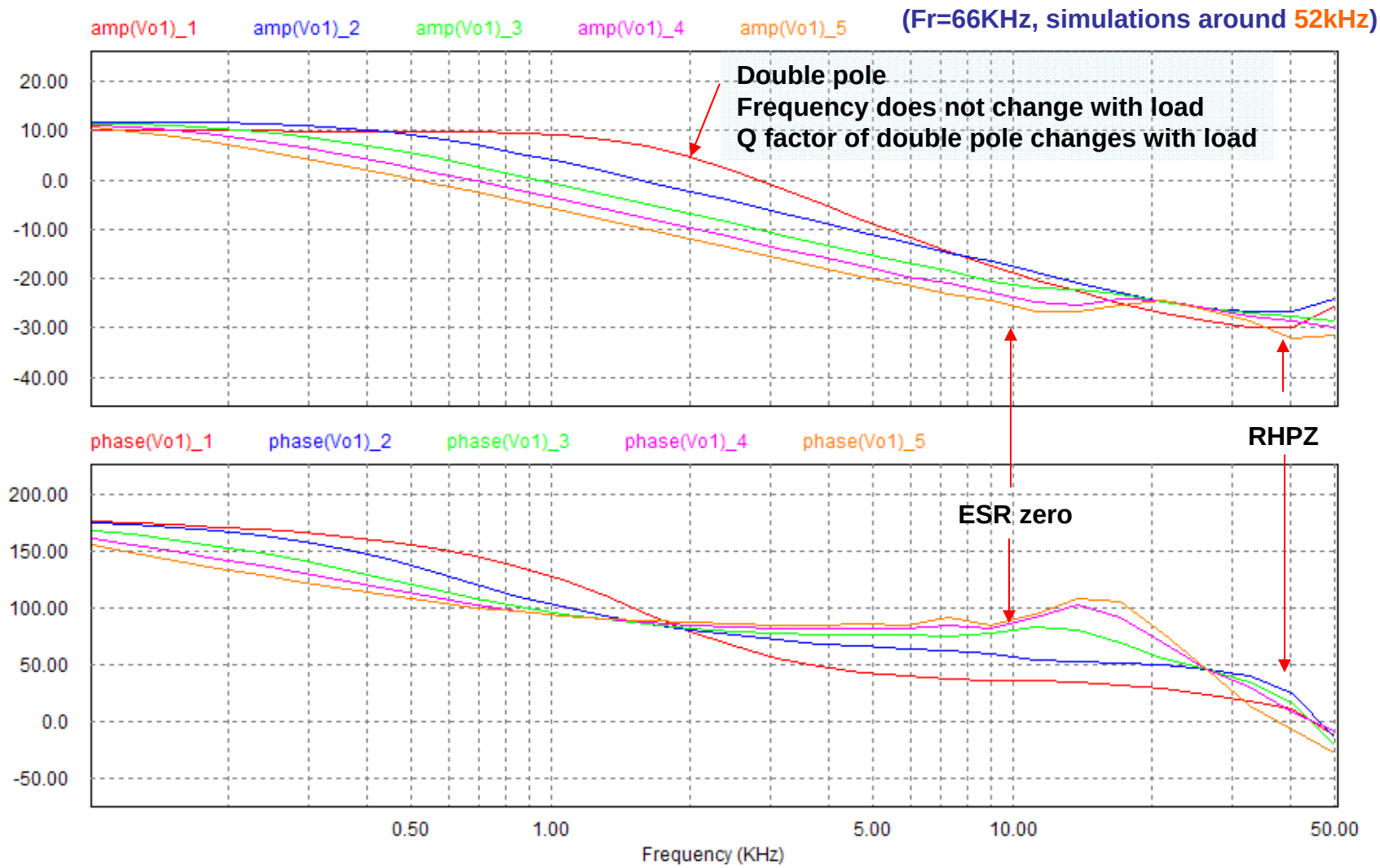
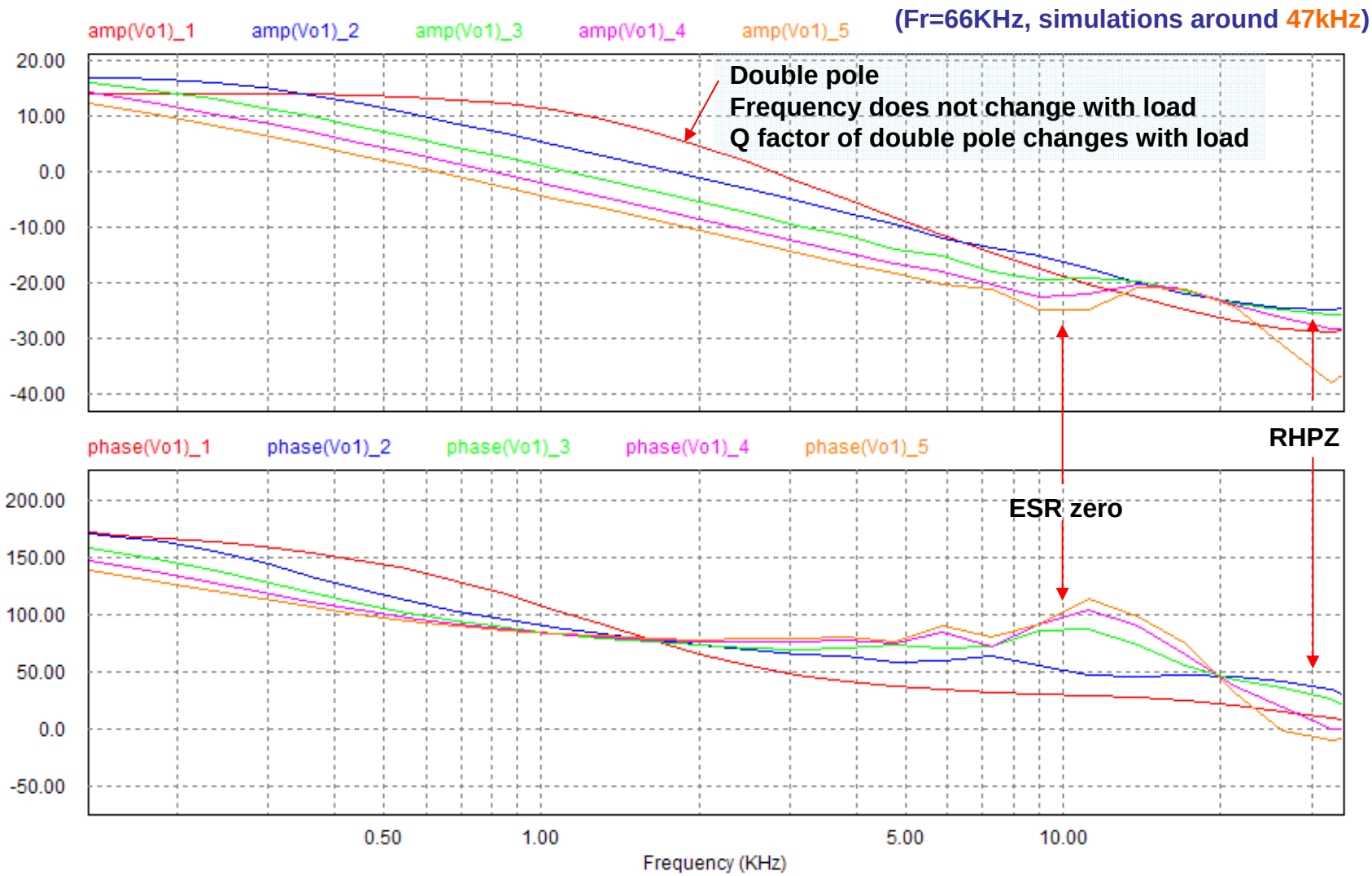
2. 一个低频极点，其位置与输出电容、负载和谐振电路参数有关当负载减小时极点的位置往低频移动
3. 差频双极点,频率在(f-fr)，当开关频率接近谐振频率时，双极点分成两个极点(低Q值双极点)，一个接近于低频极点，一个移到比较高的频率
4. 有RHPZ存在，但频率非常高，远大于一般AC/DC的带宽，不会影响补偿

低于谐振频率不同负载时的小信号传递特性1



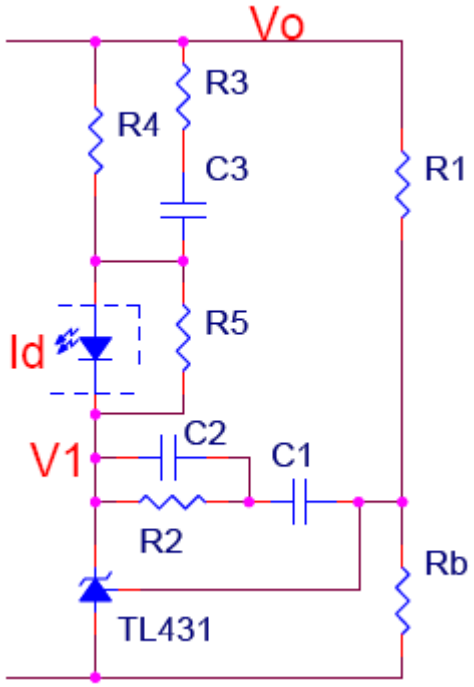
低于谐振频率不同负载时的小信号传递特性2



低于谐振频率不同负载时的小信号传递特性

1. 一个零点，有电容的ESR形成
2. 低频双极点，其位置相当稳定，随开关频率移动很小
3. 有RHPZ存在，但频率非常高，远大于一般AC/DC的带宽，不会影响补偿

反馈补偿方案



$$V1 = -\frac{\frac{1}{sC1} + \frac{1}{sC2} \| \frac{1}{R2}}{R1} * V_o$$

零点放在低频初，补偿轻载时的低频极点

零点放在双极点处，提高带宽和相位

$$\frac{I_d}{V_o} = \frac{V_o - V1}{R4 \| \left(R3 + \frac{1}{sC3} \right)} / V_o \approx \frac{[1 + s(R1 + R2)C1] * [1 + s(R3 + R4)C3]}{sR1R4C1(1 + sR2C2)(1 + sR3C3)}$$

3型补偿网络

低频极点，提高低频增益(R4)

衰减开关噪音

抵消ESR零点

一款新颖的控制芯片

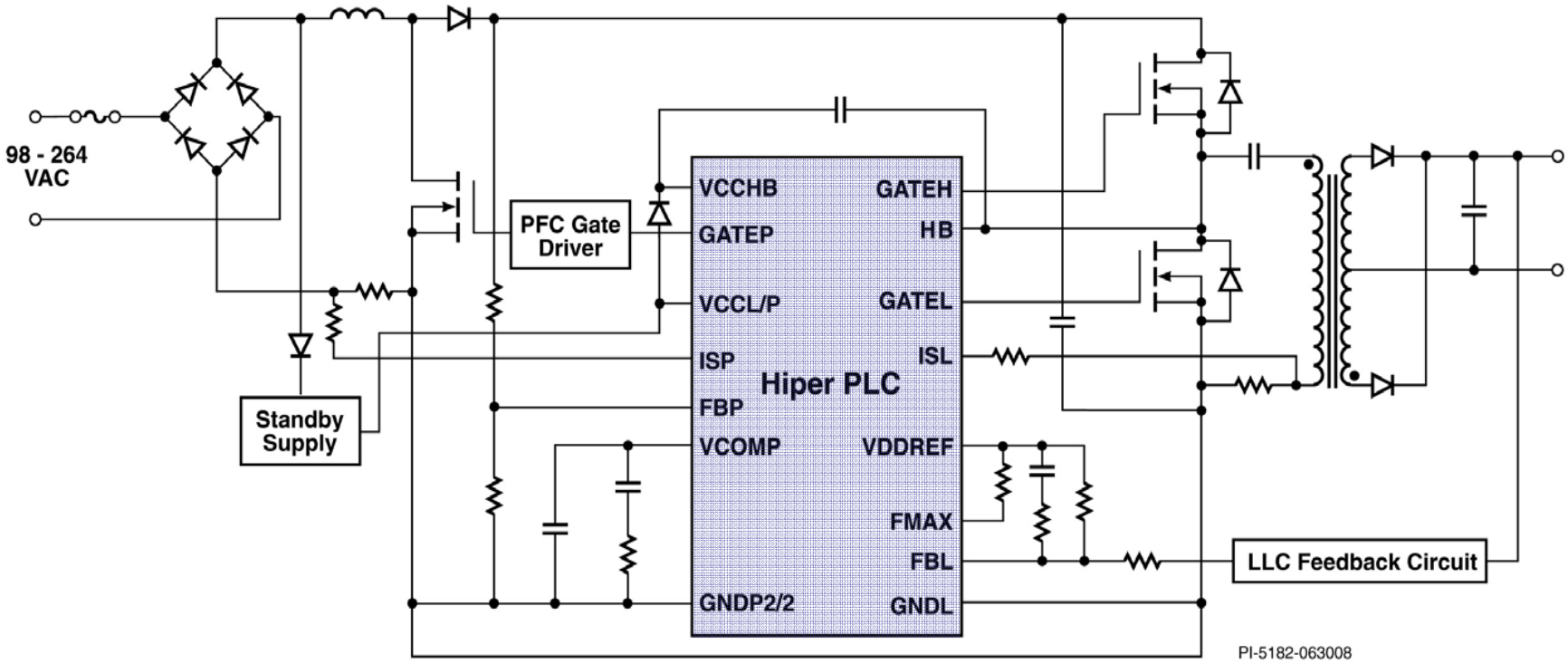
POWER INTEGRATIONS®

HiperPLC

用于LCD TV的全新高集成解决方案

HiperPLC：LCD TV主电源解决方案

- 集成度最高的LCD TV，LED路灯主电源解决方案
  - 集成了PFC、LLC控制器以及LLC半桥驱动器
- 非常适合150 W至600 W的功率应用



主要特点及优点

1. 连续模式PFC，Doff 控制无需检测输入波形

   CCM模式降低了差模干扰和磁损，Doff控制减少了外部元件

2. PFC与LLC级之间的频率相位锁定

   避免差频干扰，纹波电流抵消降低了高压电解成本

3. PFC/LLC避免边缘冲突

   避免了互相干扰

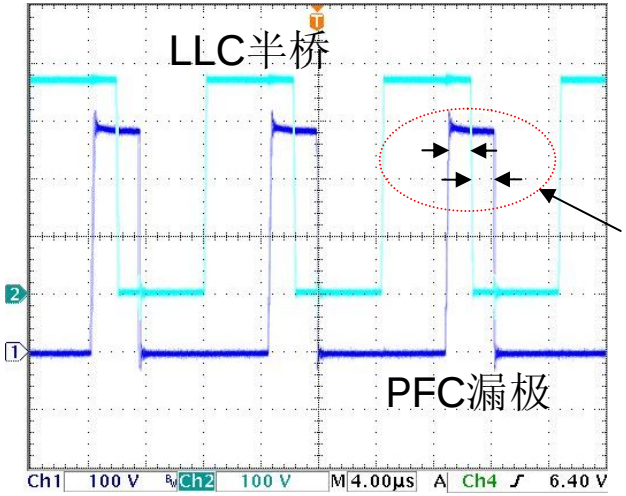
4. 占空比匹配与严格限制的死区时间优化了LLC效率

   两相不平衡度小于1％

5. 集成了供电控制和各种保护功能

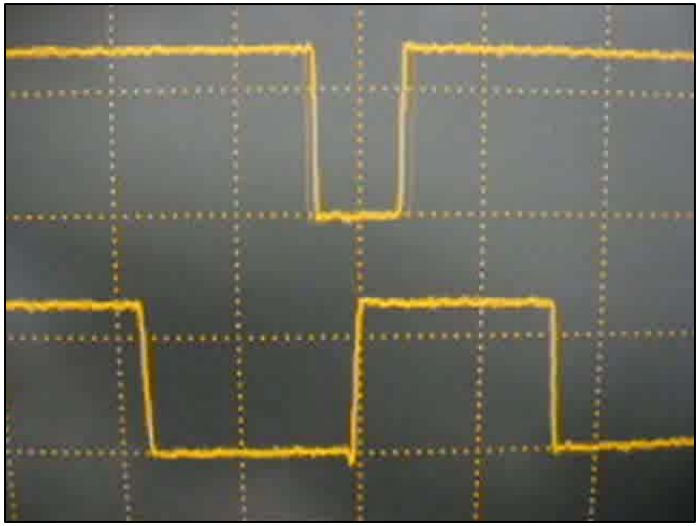
HiperPLC LLC/PFC同步可节省成本

- LLC与PFC之间无差频
- 纹波抵消
- 无内部边缘冲突
  - 边缘从不相交，这样可以降低噪声及互相干扰
- 简化单层电路板的布局



PFC围绕LLC边缘对称分布

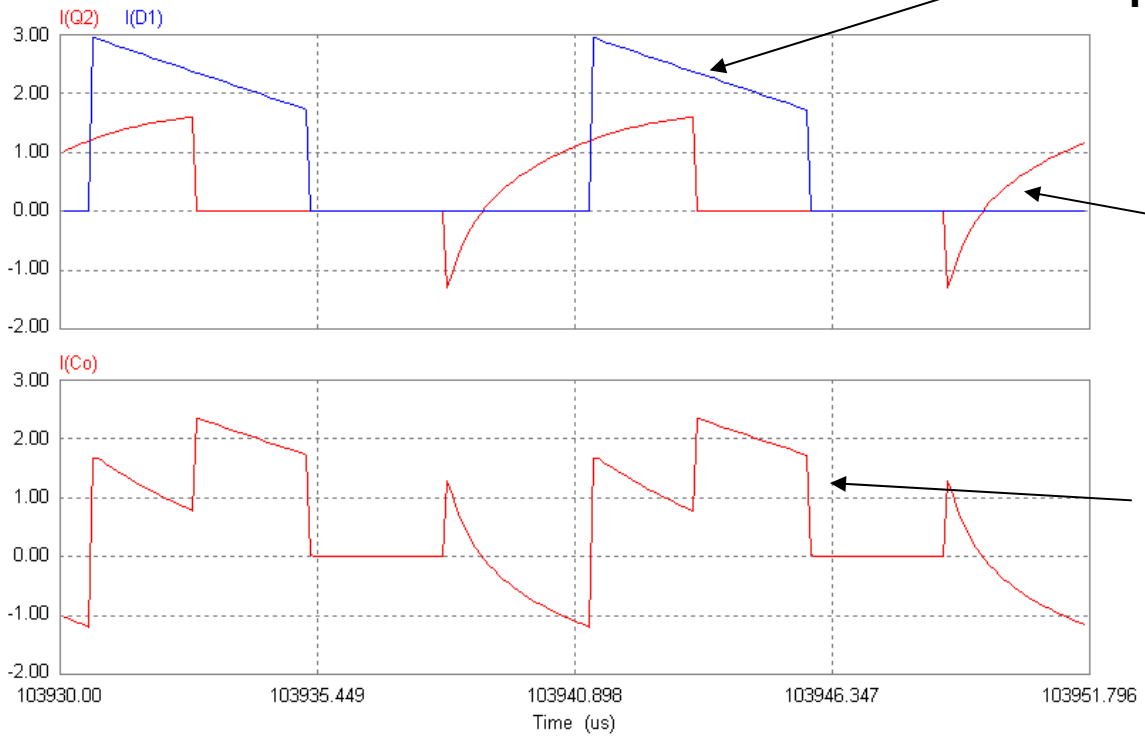
HiperPLC – 显示LLC与PFC相位锁定



实测波形

PFC & LLC 仿真

纹波减少 (Vac(max), 接近正弦)

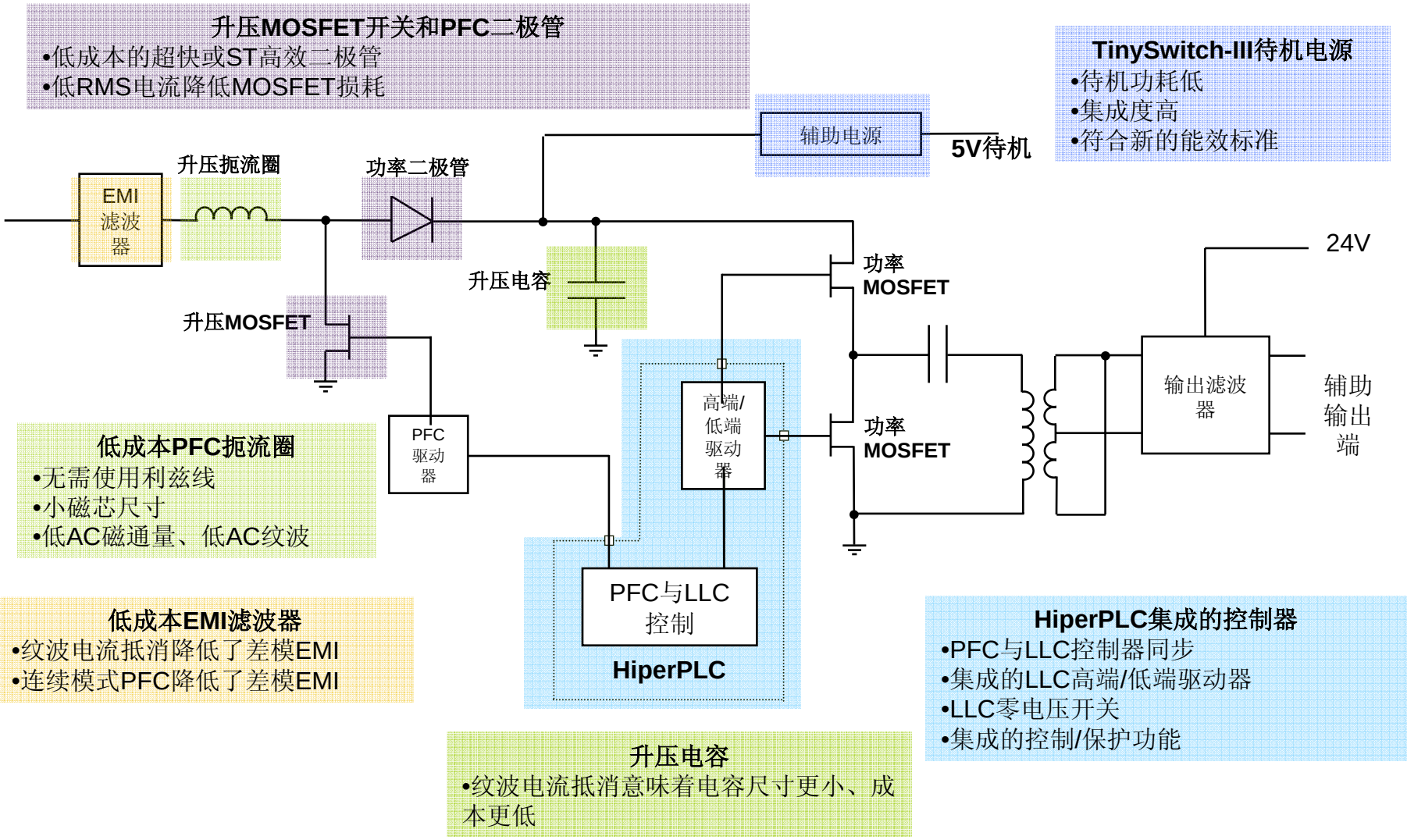
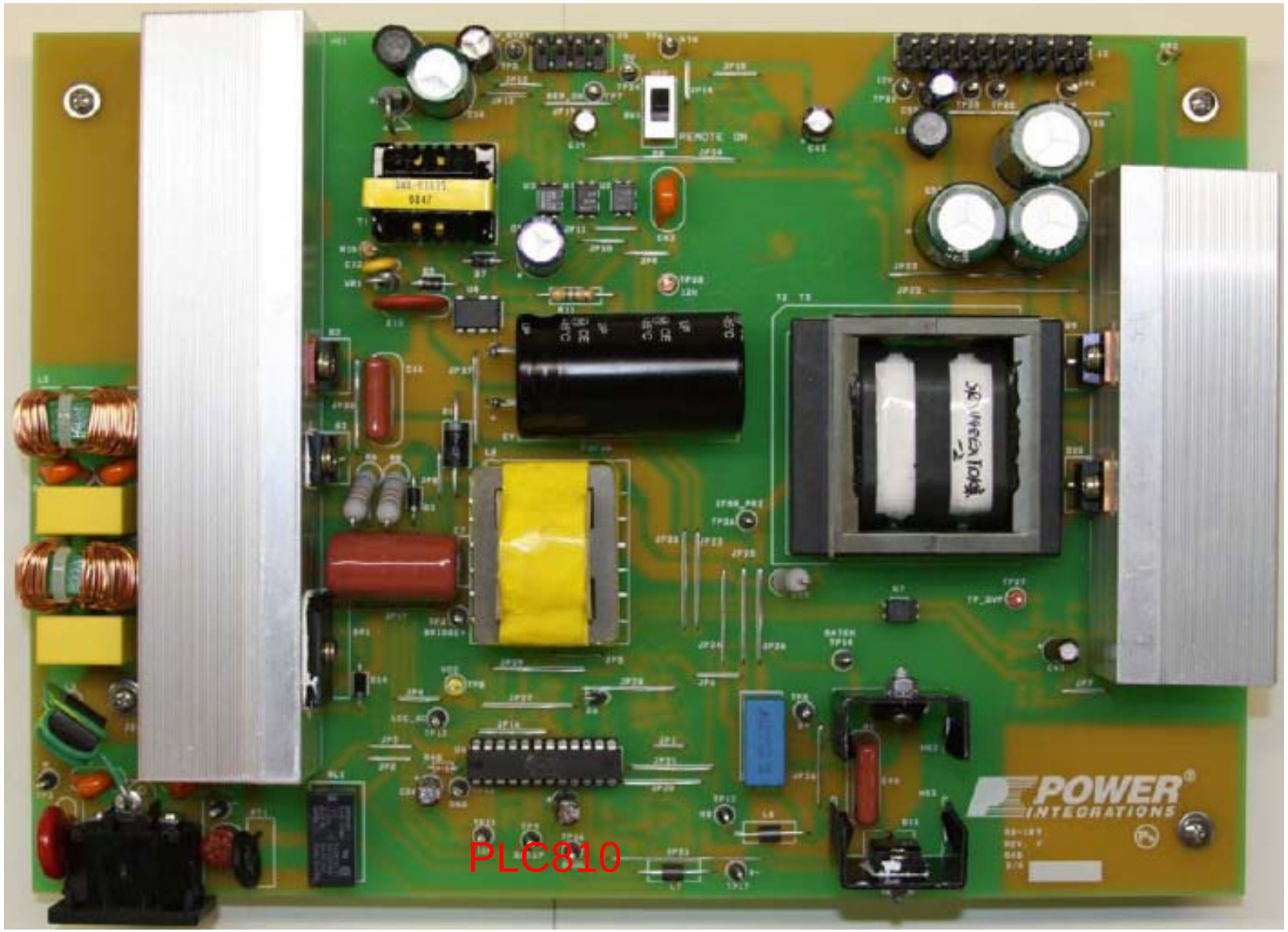


PFC 电流流入 **PFC** 输出电容

LLC 流出 **PFC** 输出电容

PFC 输出电容的电流减少 **(**电流抵消**)**

参考设计结构框图


升压MOSFET开关和PFC二极管
•低成本的超快或ST高效二极管
•低RMS电流降低MOSFET损耗
TinySwitch-III待机电源
•待机功耗低
•集成度高
•符合新的能效标准
辅助电源
5V待机
升压扼流圈
功率二极管
EMI
滤波
器
升压电容
升压MOSFET
功率
MOSFET
24V
输出滤波
器
辅助
输出
端
PFC
驱动
器
高端/
低端
驱动
器
功率
MOSFET
低成本PFC扼流圈
•无需使用利兹线
•小磁芯尺寸
•低AC磁通量、低AC纹波
PFC与LLC
控制
HiperPLC
低成本EMI滤波器
•纹波电流抵消降低了差模EMI
•连续模式PFC降低了差模EMI
HiperPLC集成的控制器
•PFC与LLC控制器同步
•集成的LLC高端/低端驱动器
•LLC零电压开关
•集成的控制/保护功能
升压电容
•纹波电流抵消意味着电容尺寸更小、成本更低

参考样板